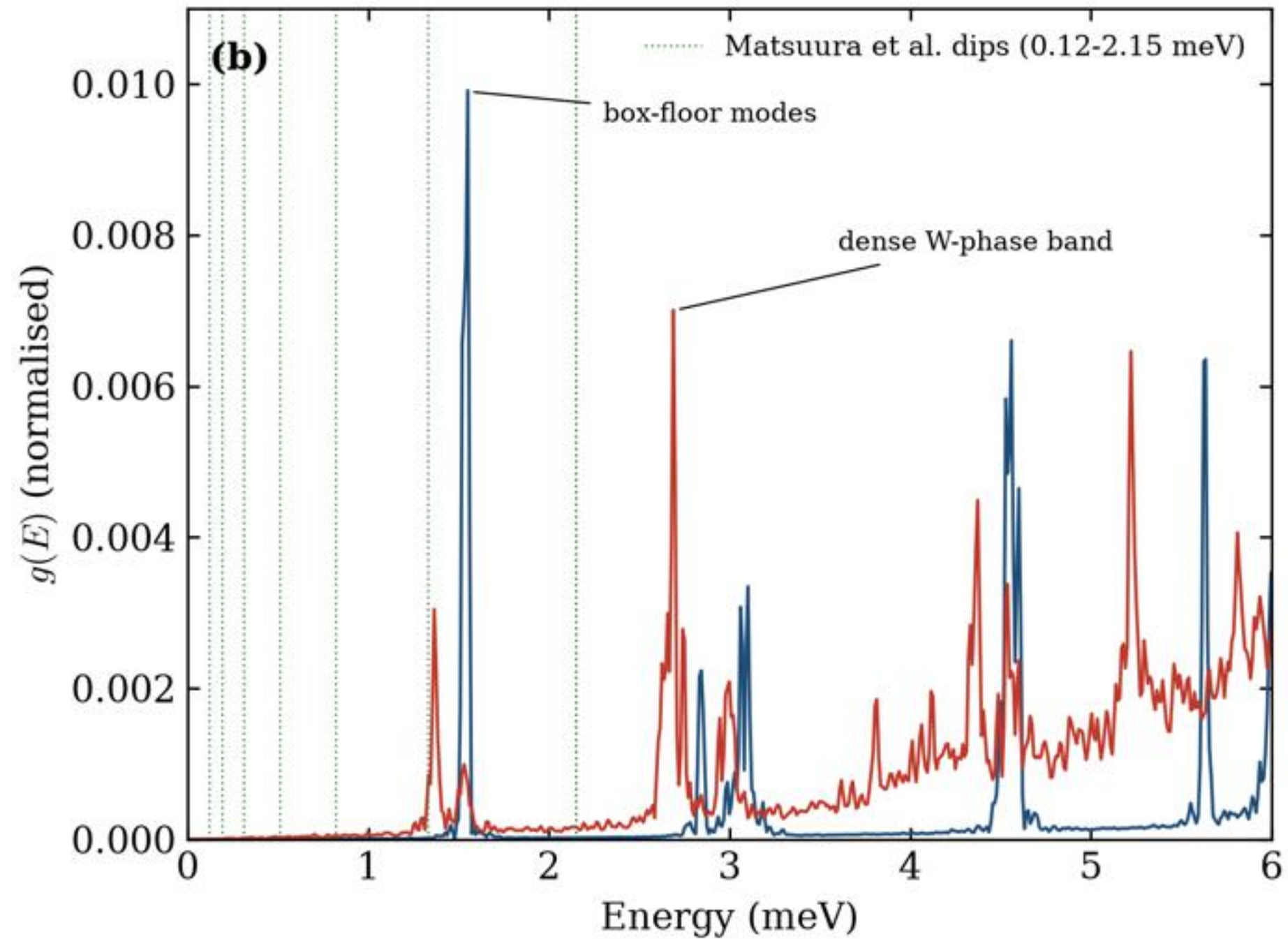
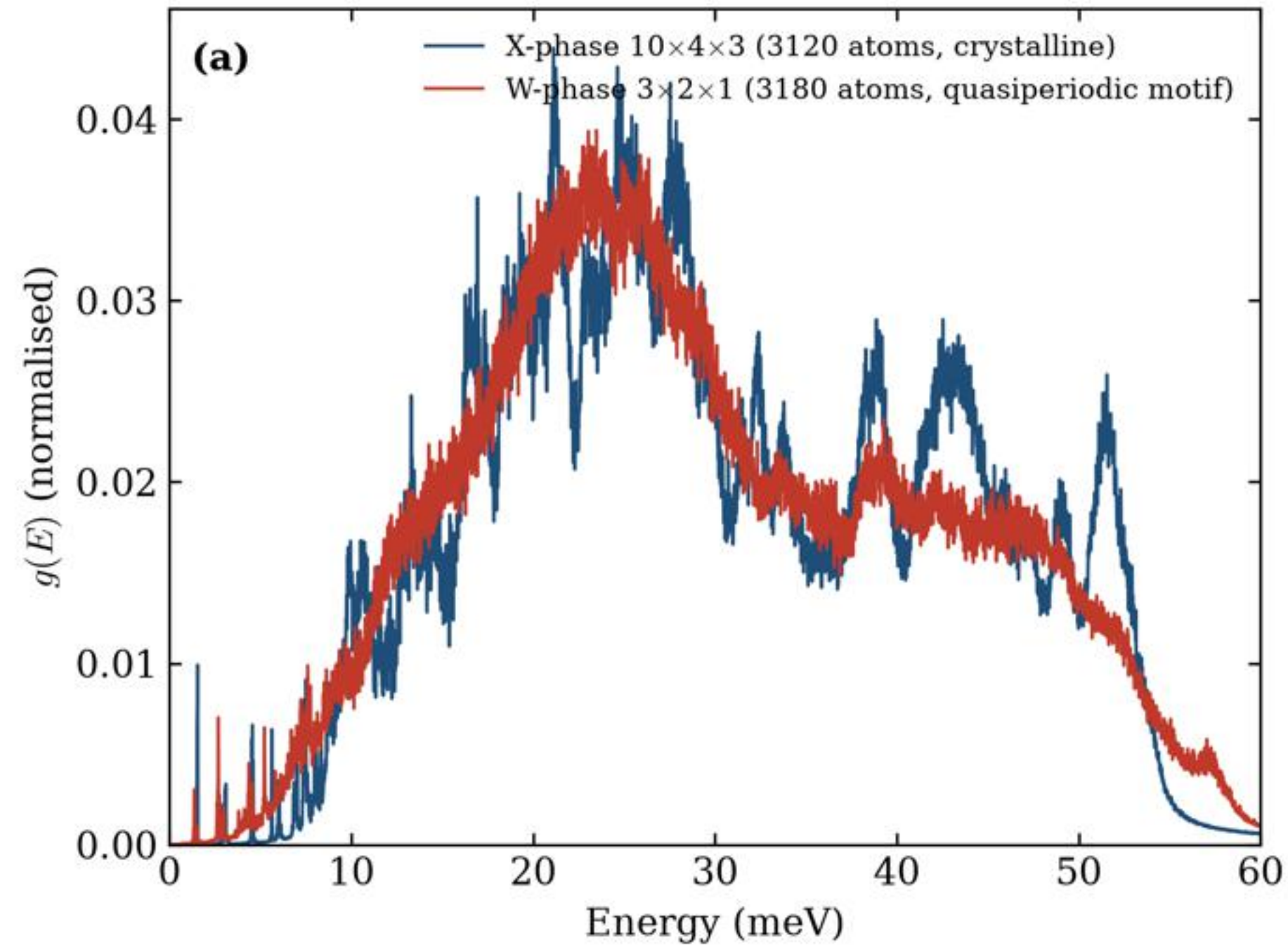
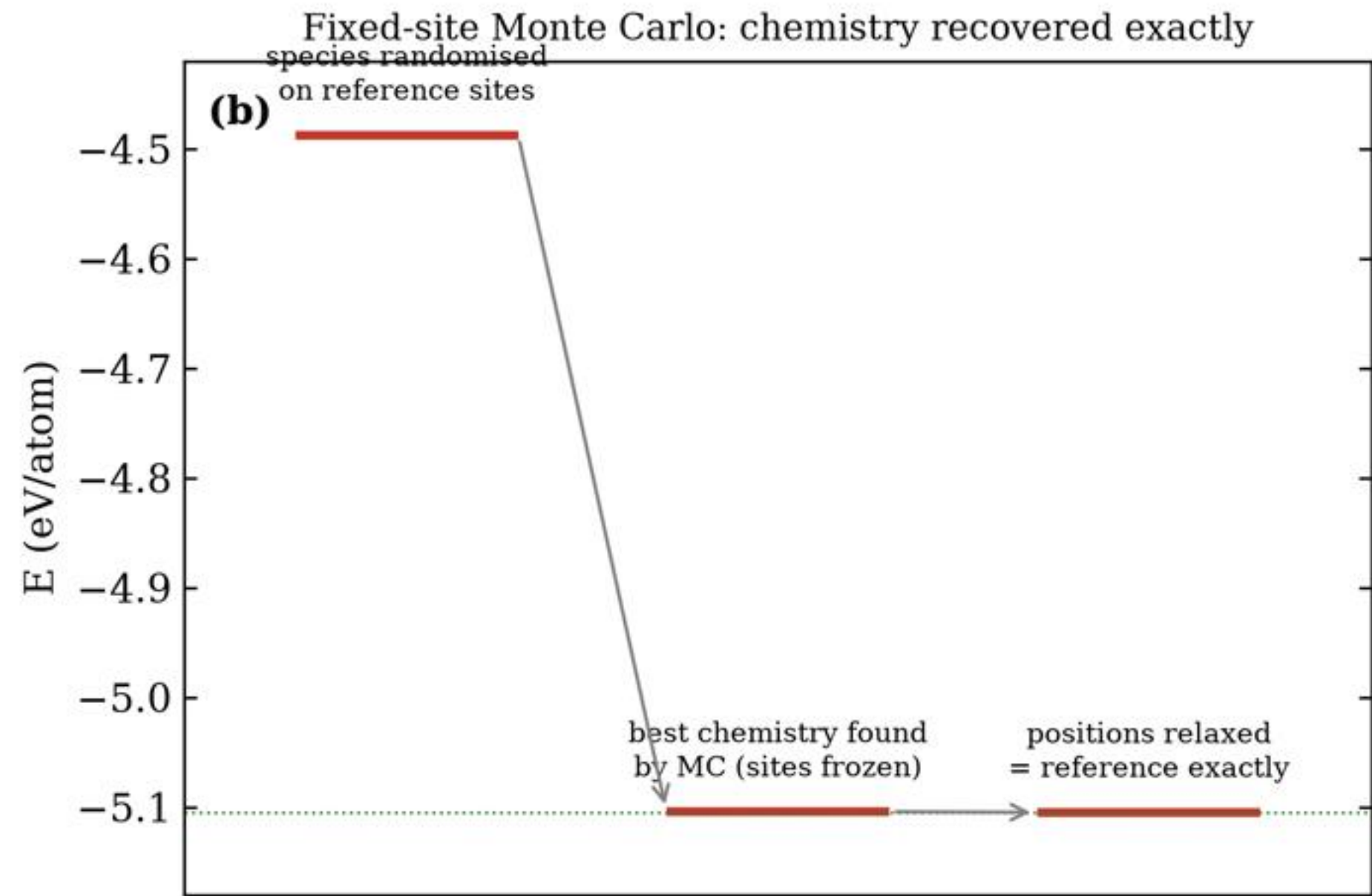
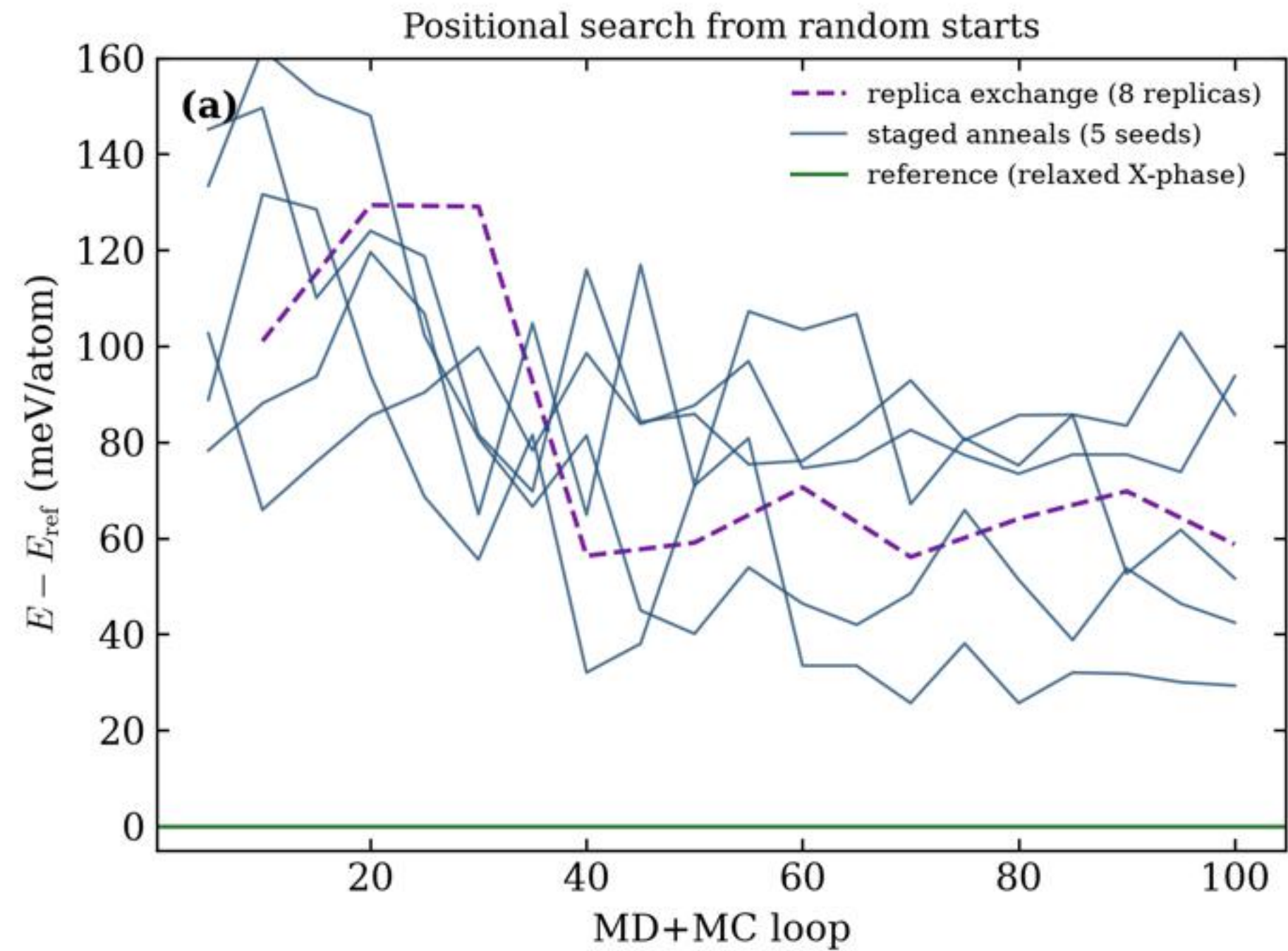
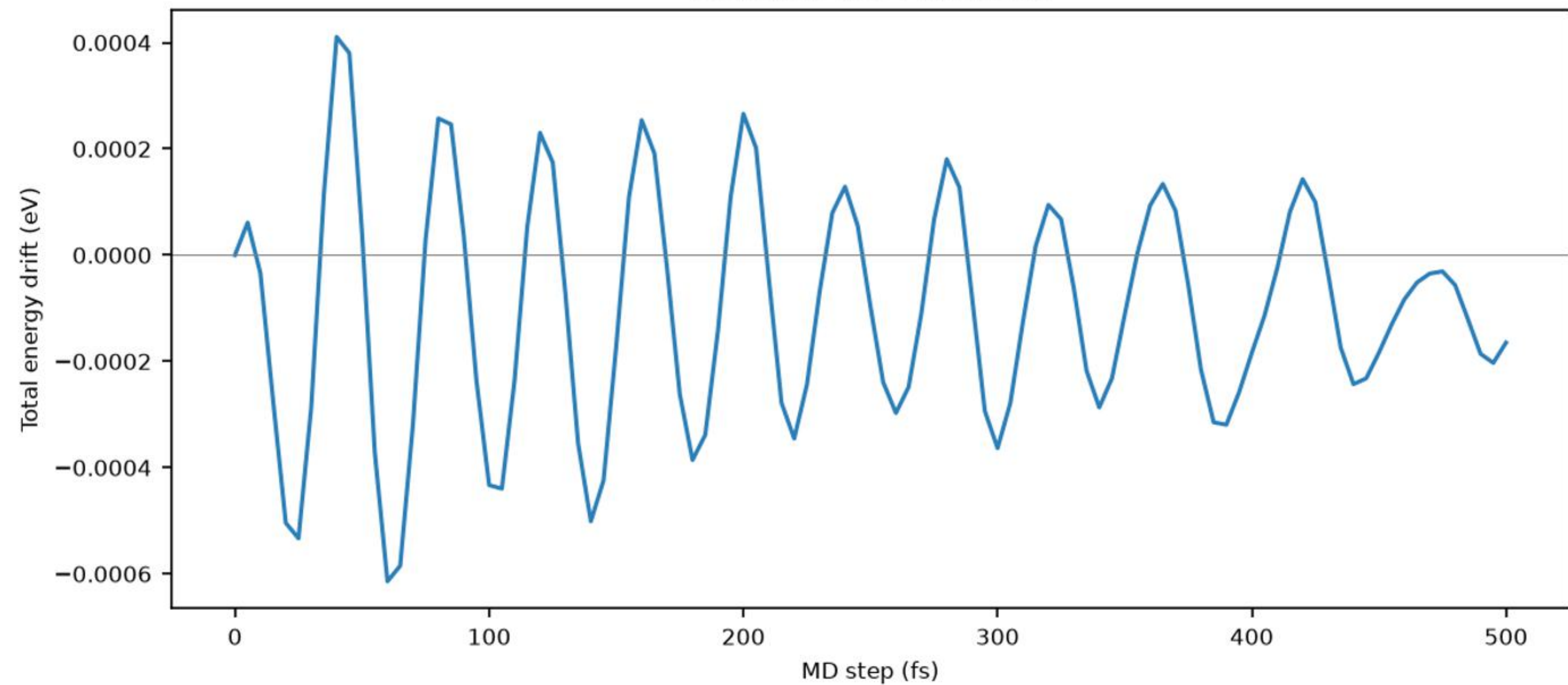




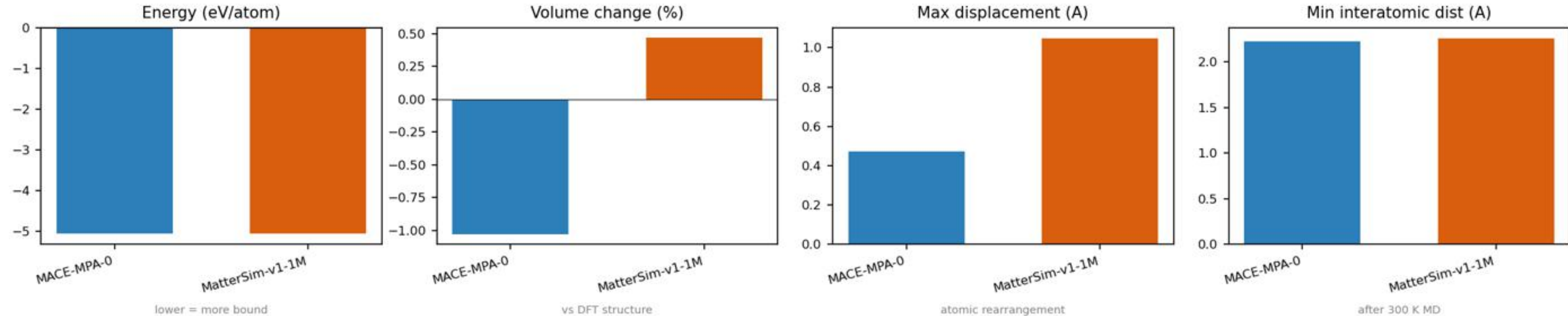


NVE energy conservation, W-AlCoNi 265 atoms

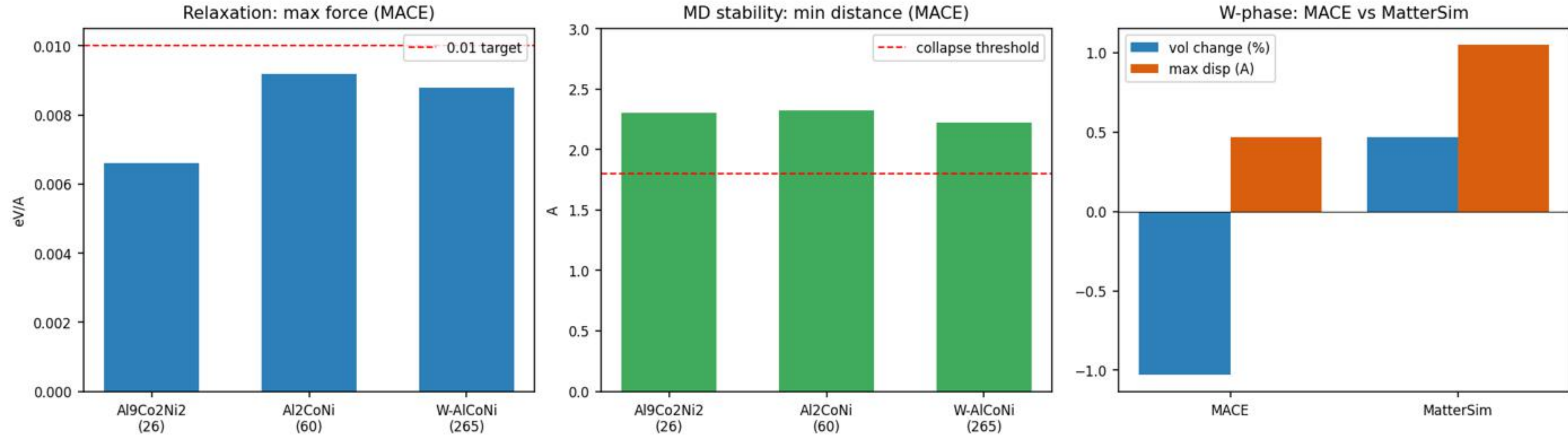
Total drift: 0.004 meV/atom



O1: MACE vs MatterSim on the 265-atom W-AlCoNi approximant

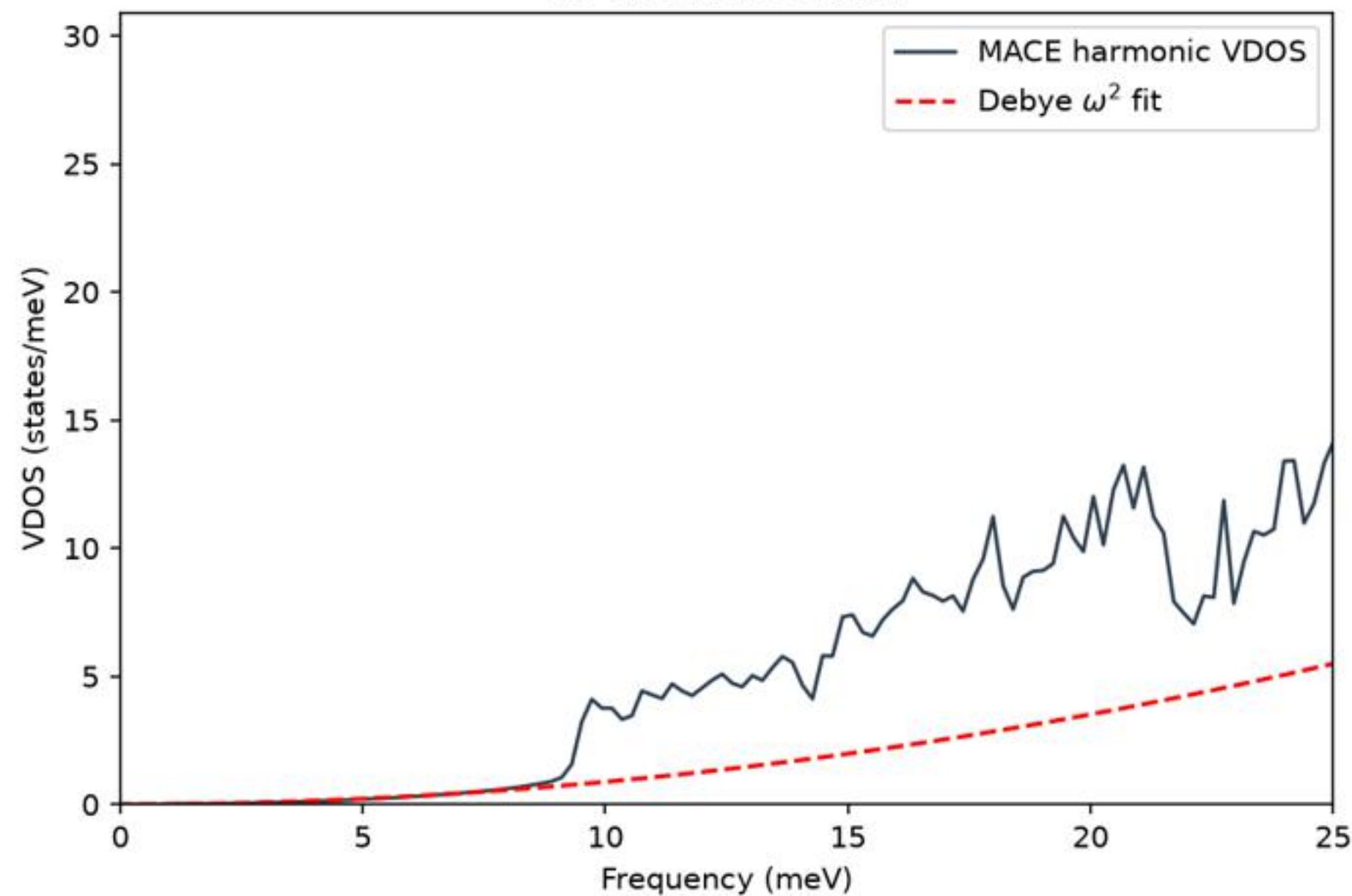


O1: MACE stability across size series, plus model comparison on W-AlCoNi

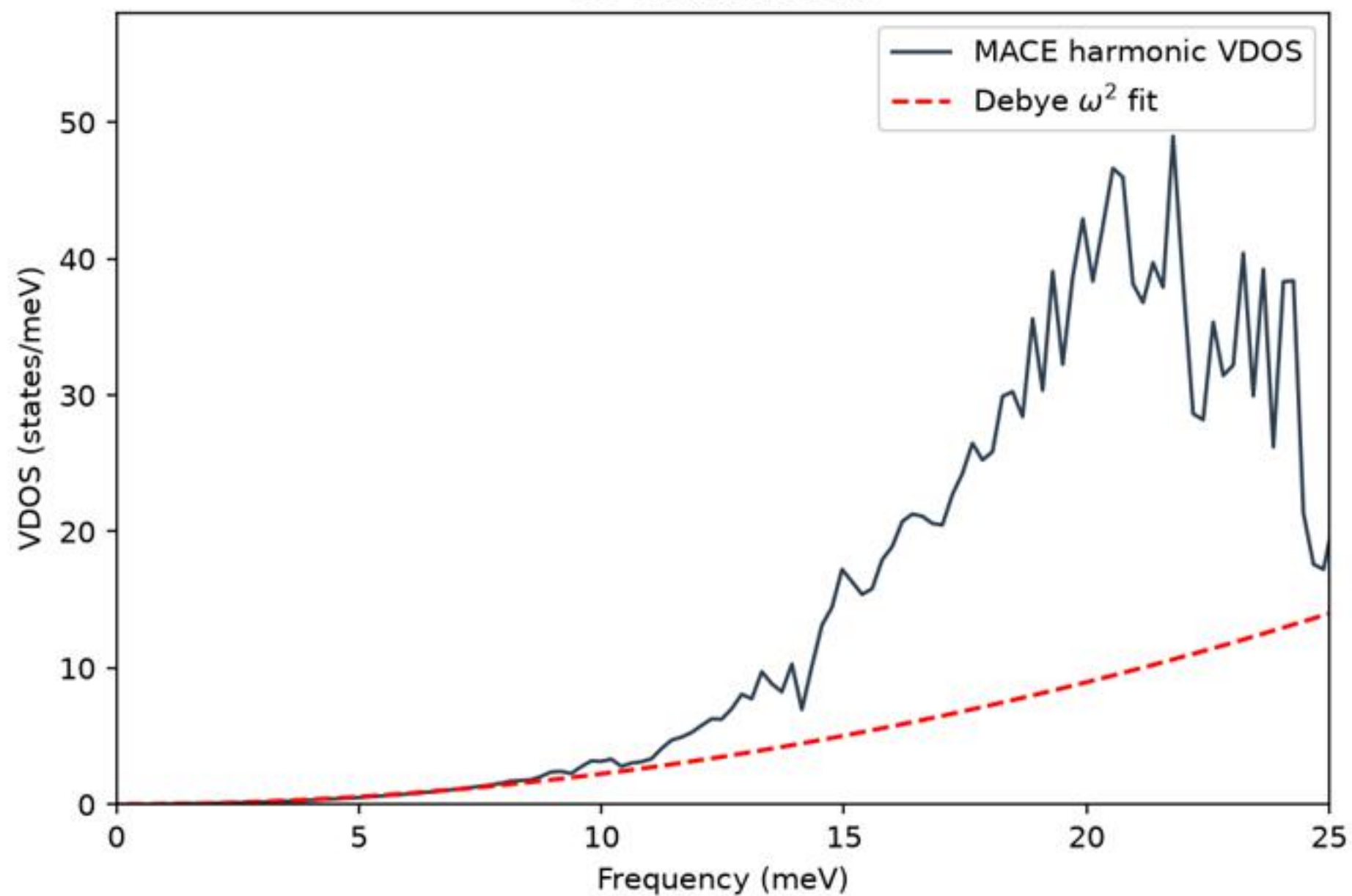


Low-frequency VDOS vs Debye prediction (Suck & Mihalkovic comparison)

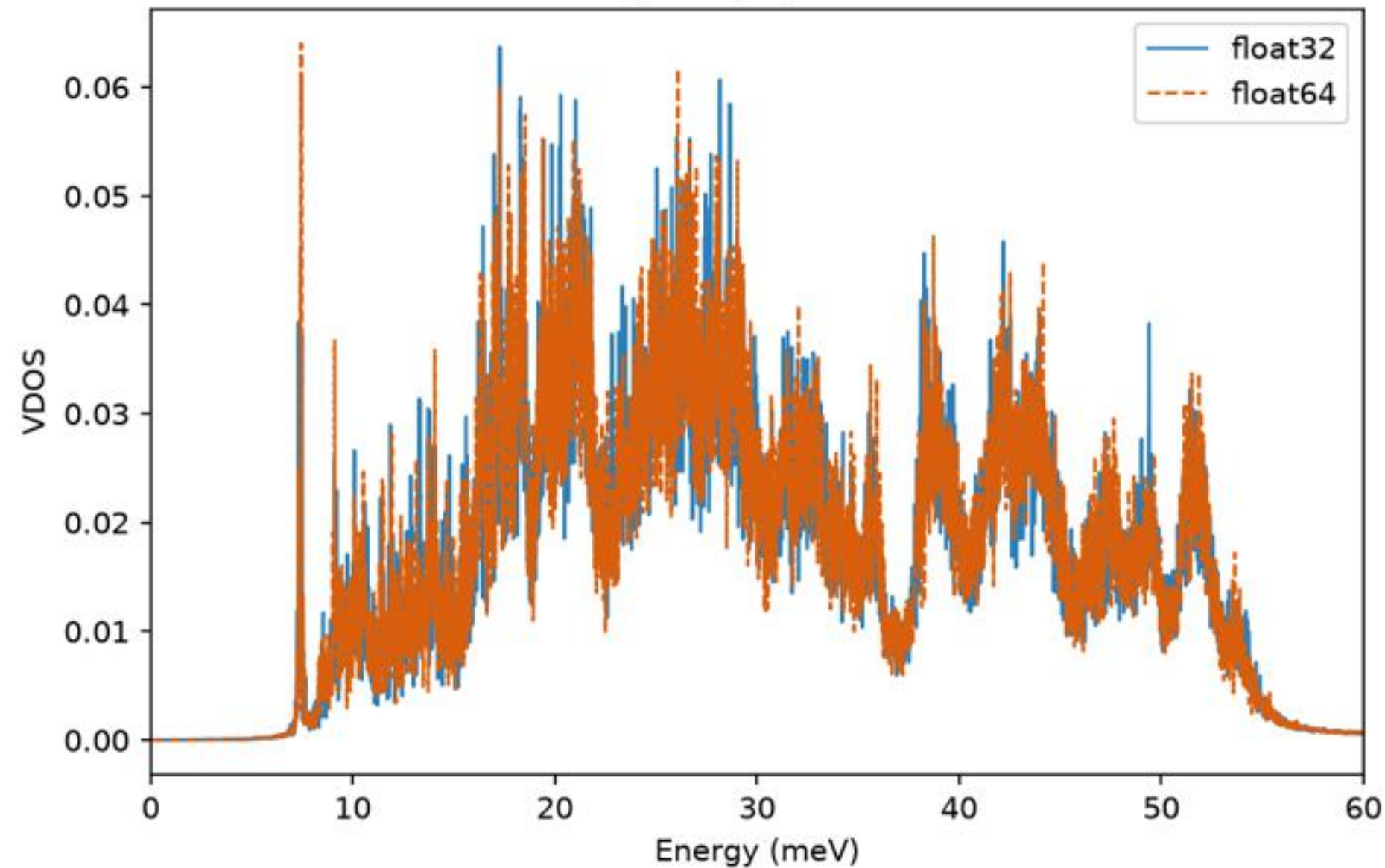
26-atom Al₉Co₂Ni₂



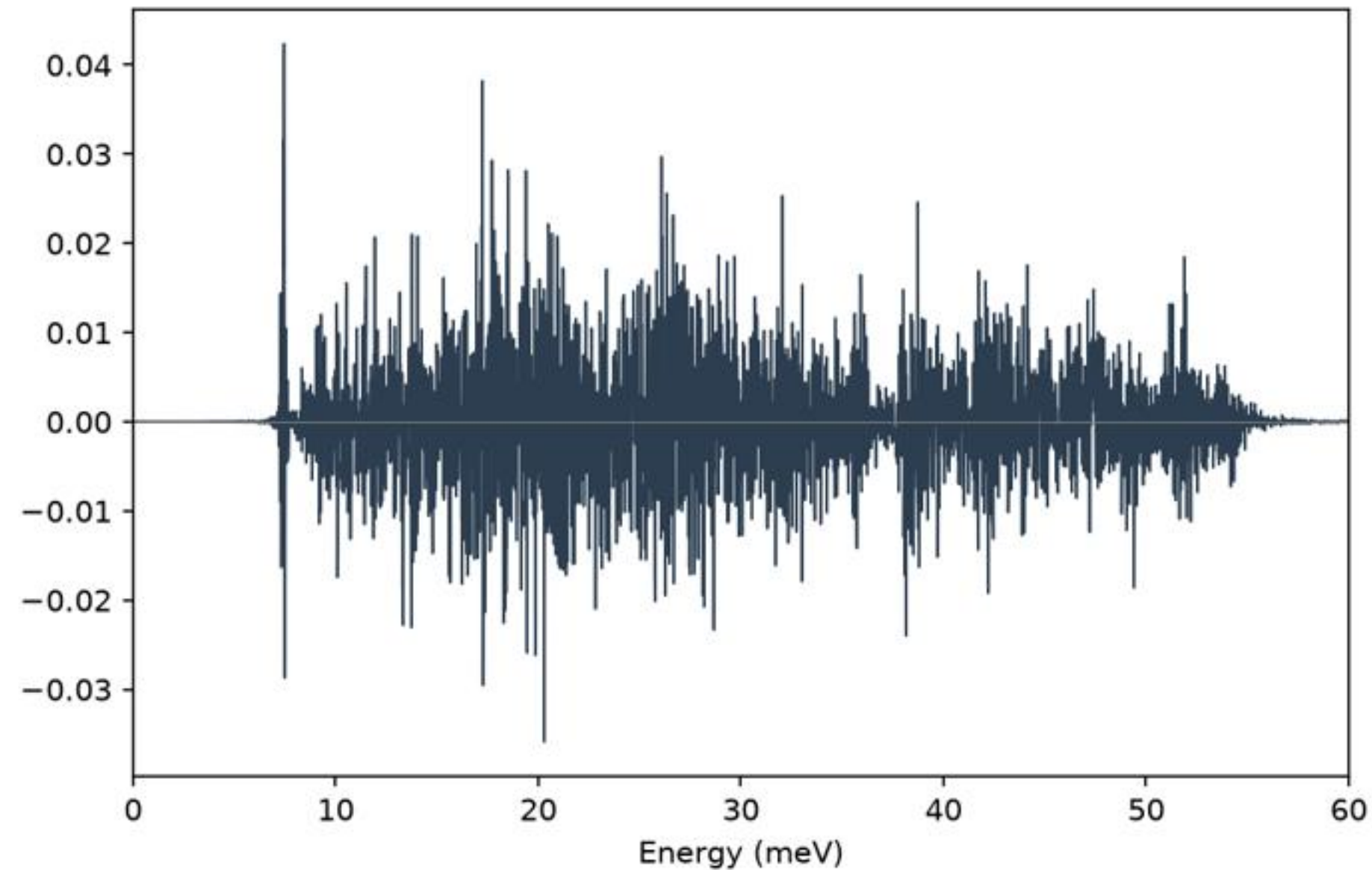
60-atom Al₂CoNi



2x2x2, 200 ps, same seed

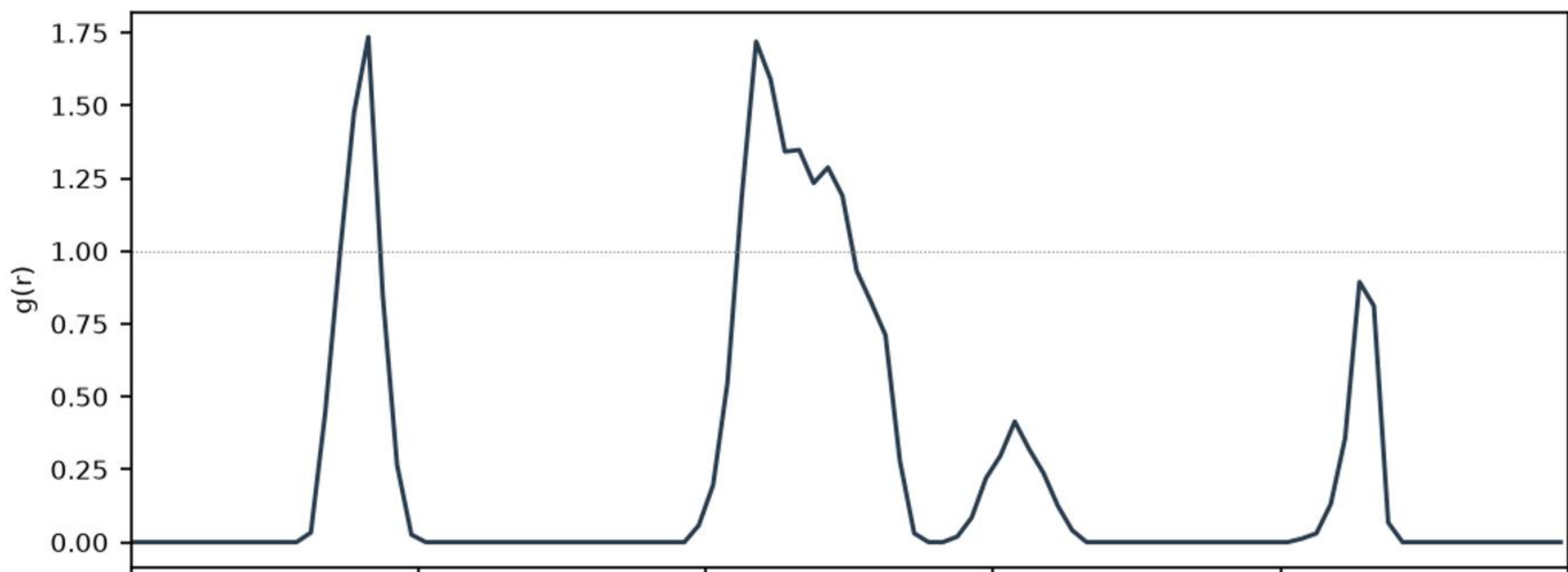


f64 - f32

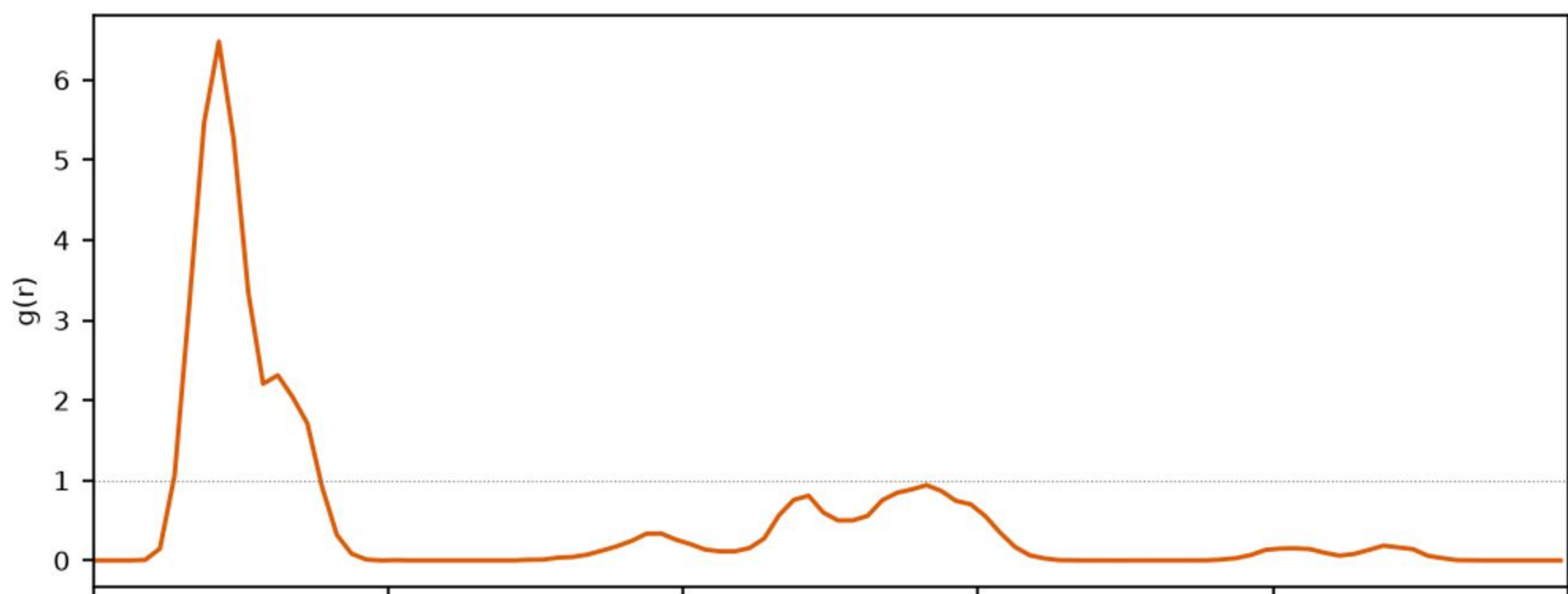


Partial pair distribution functions, 26-atom Al₉Co₂Ni₂, MD 300 K

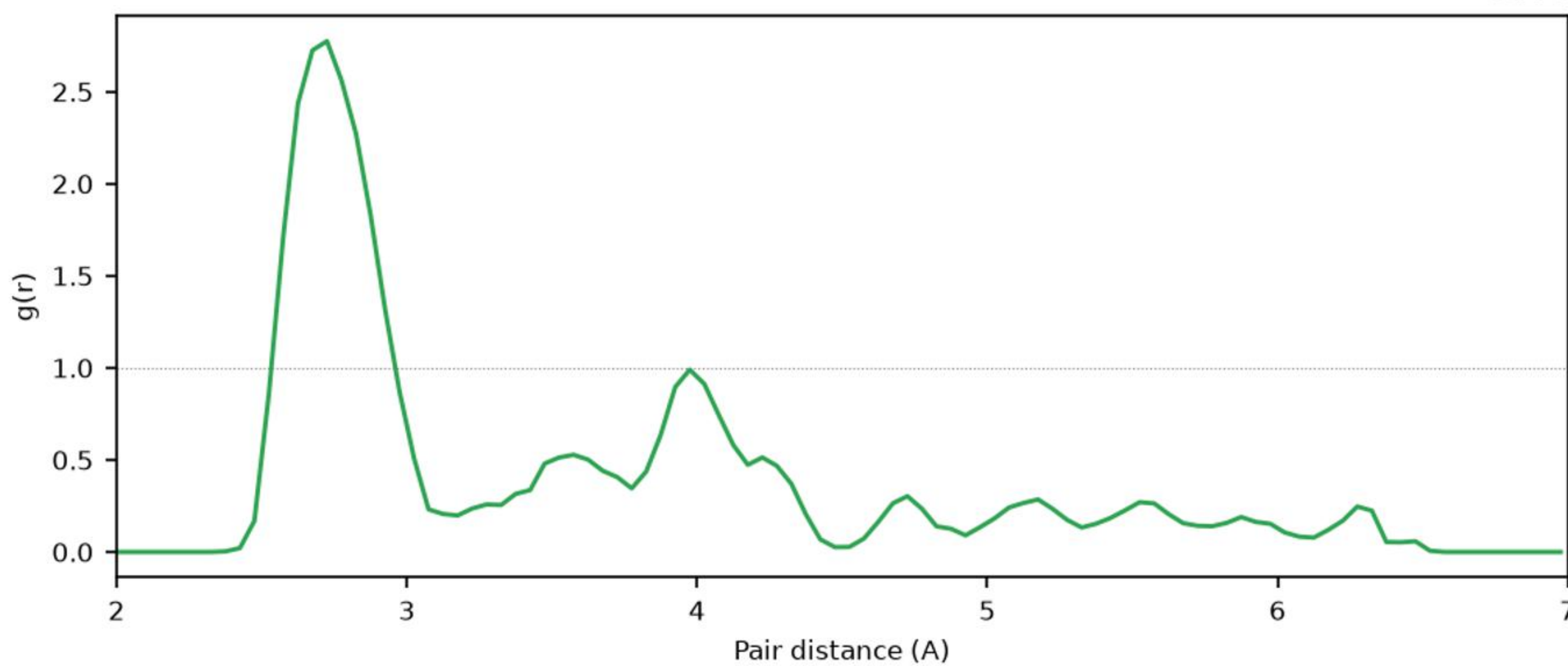
Co-Co



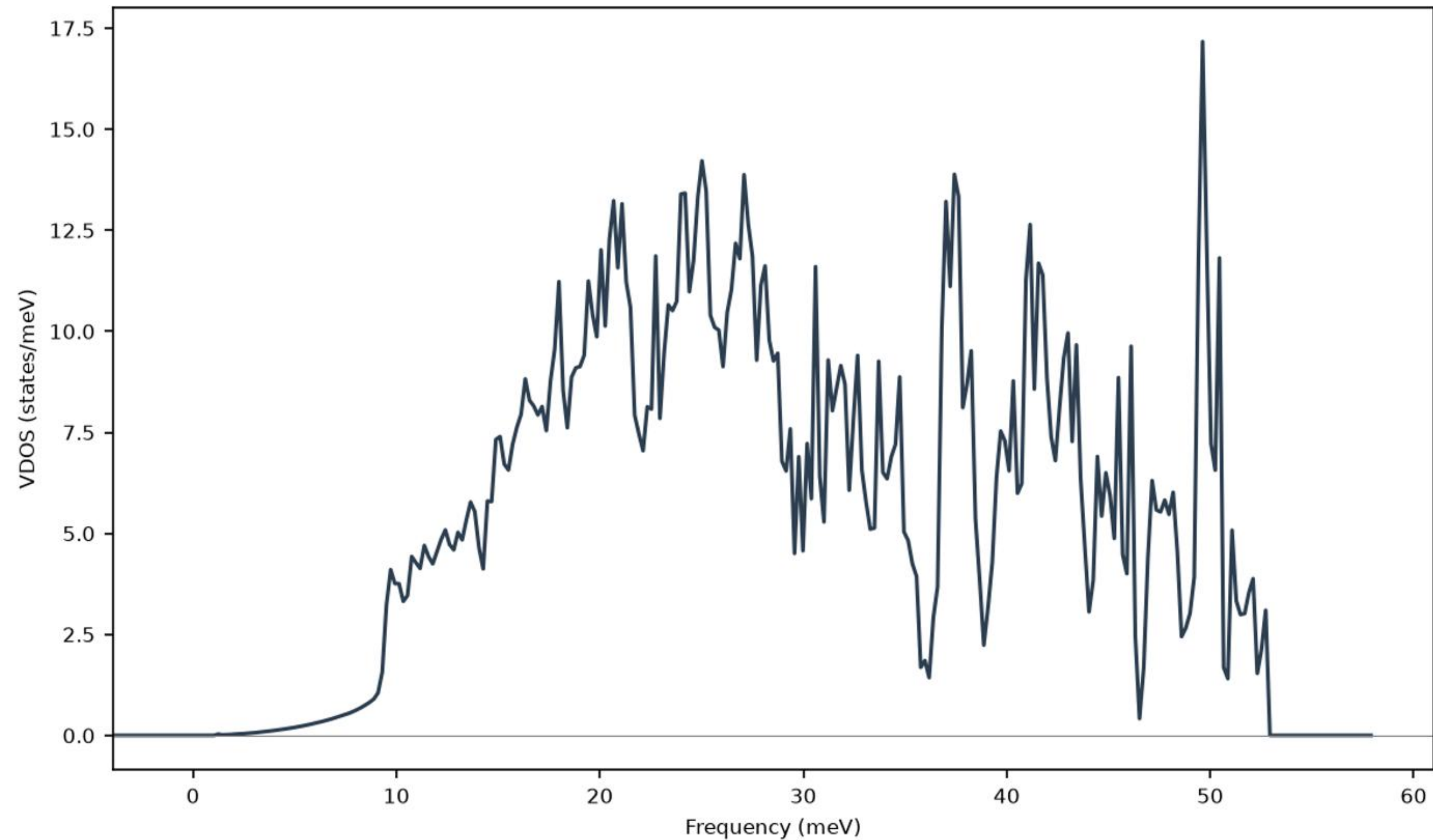
Al-Co



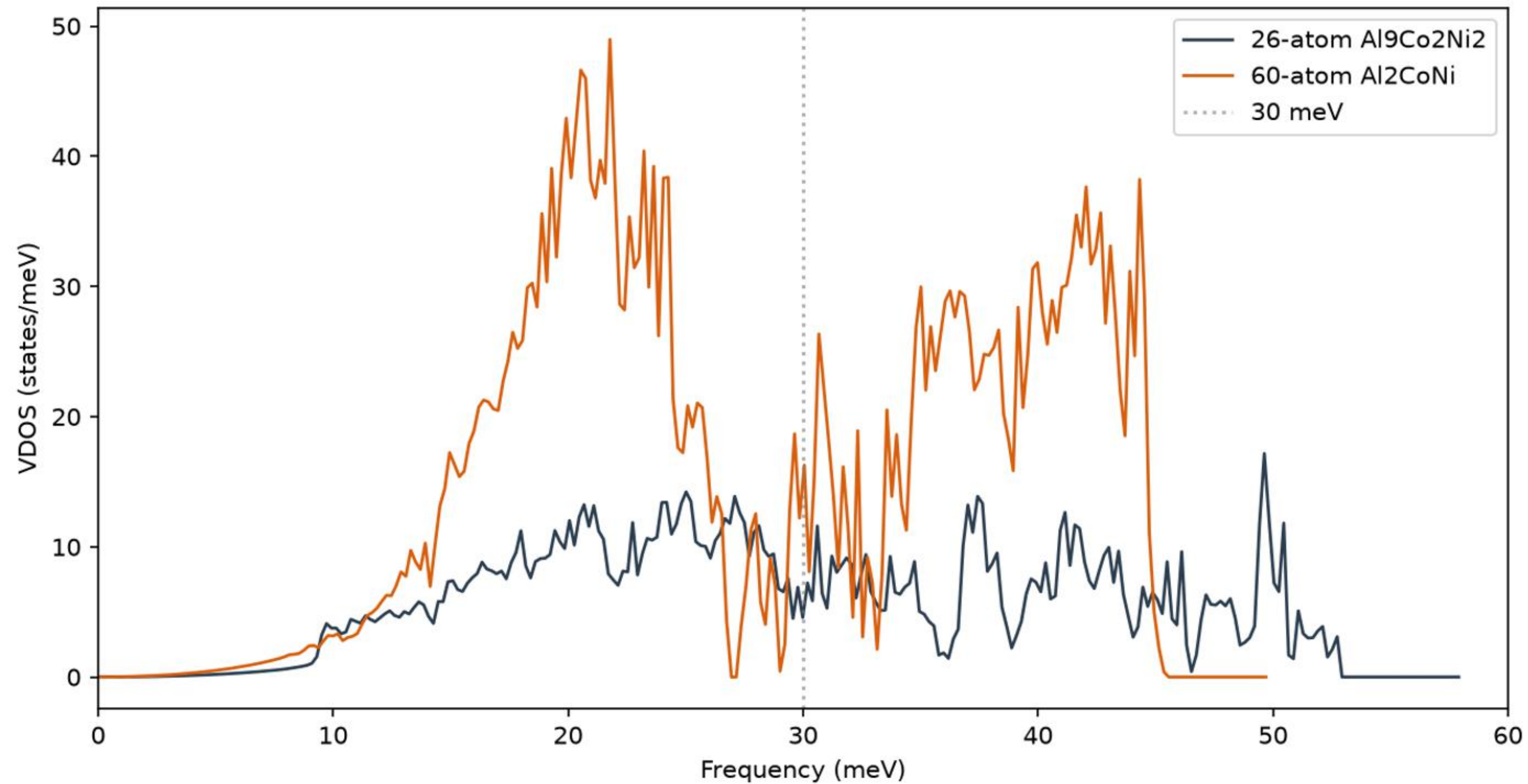
Al-Al



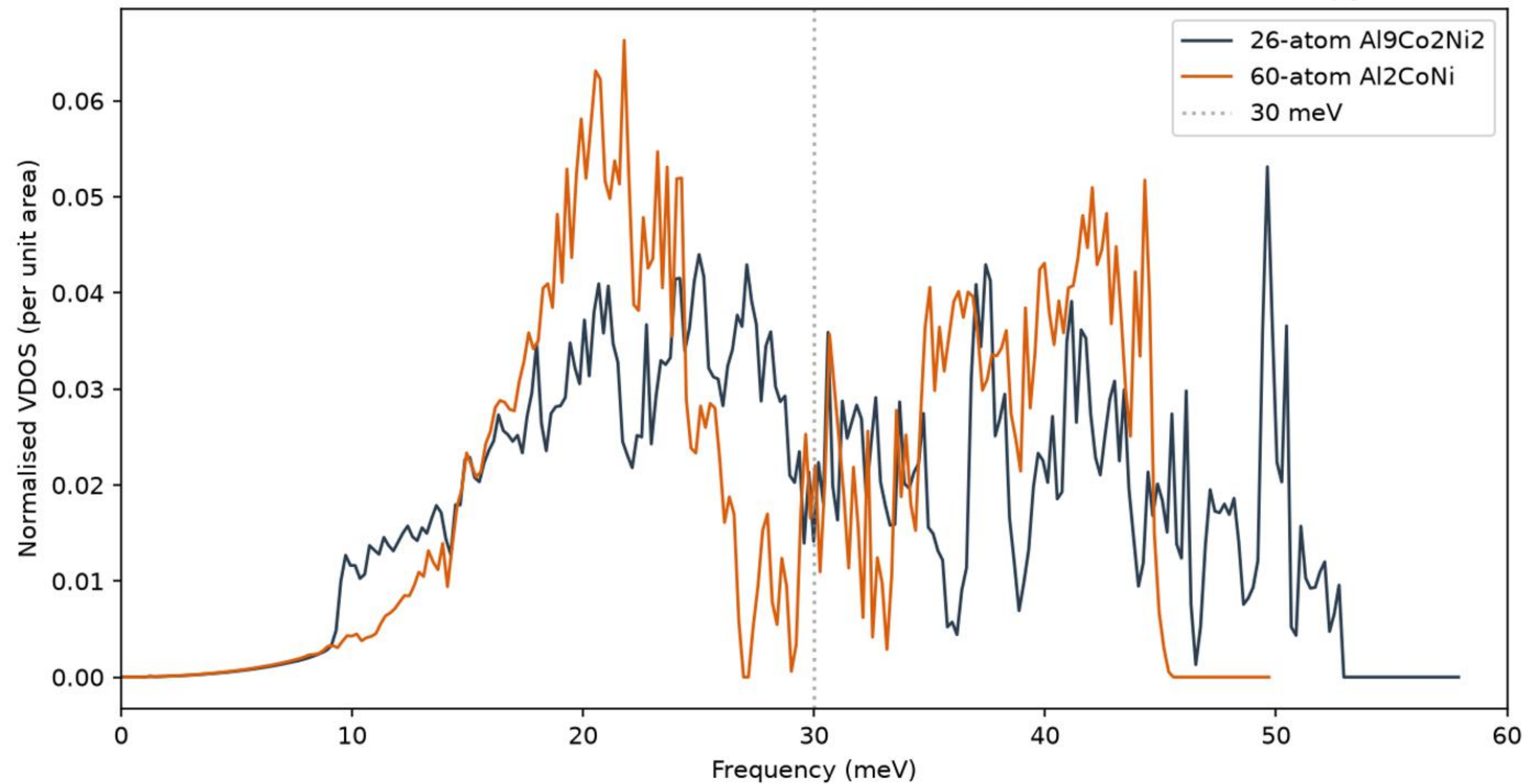
Harmonic VDOS: Al₉Co₂Ni₂ (26-atom), MACE-MPA-0 + Phonopy



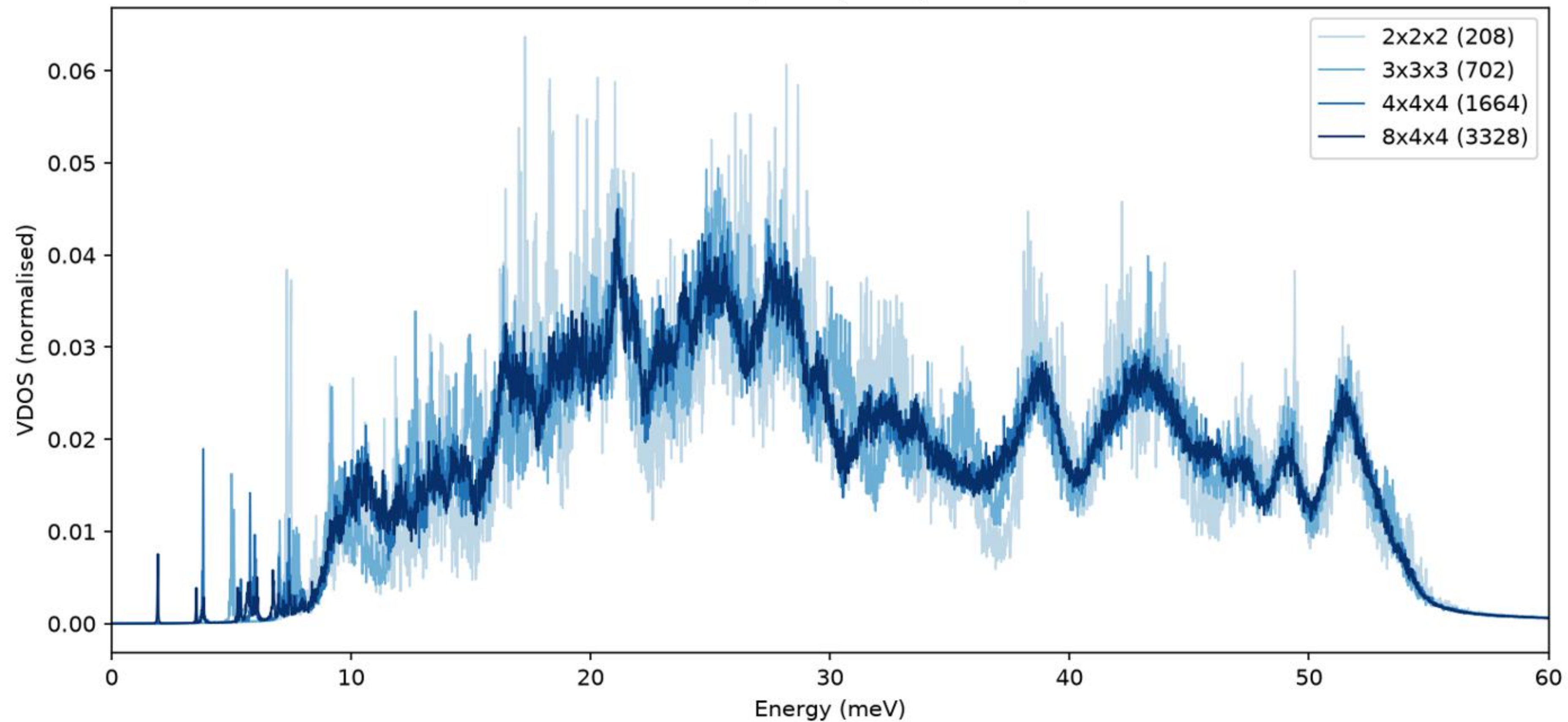
Harmonic VDOS across the size series, MACE-MPA-0 + Phonopy



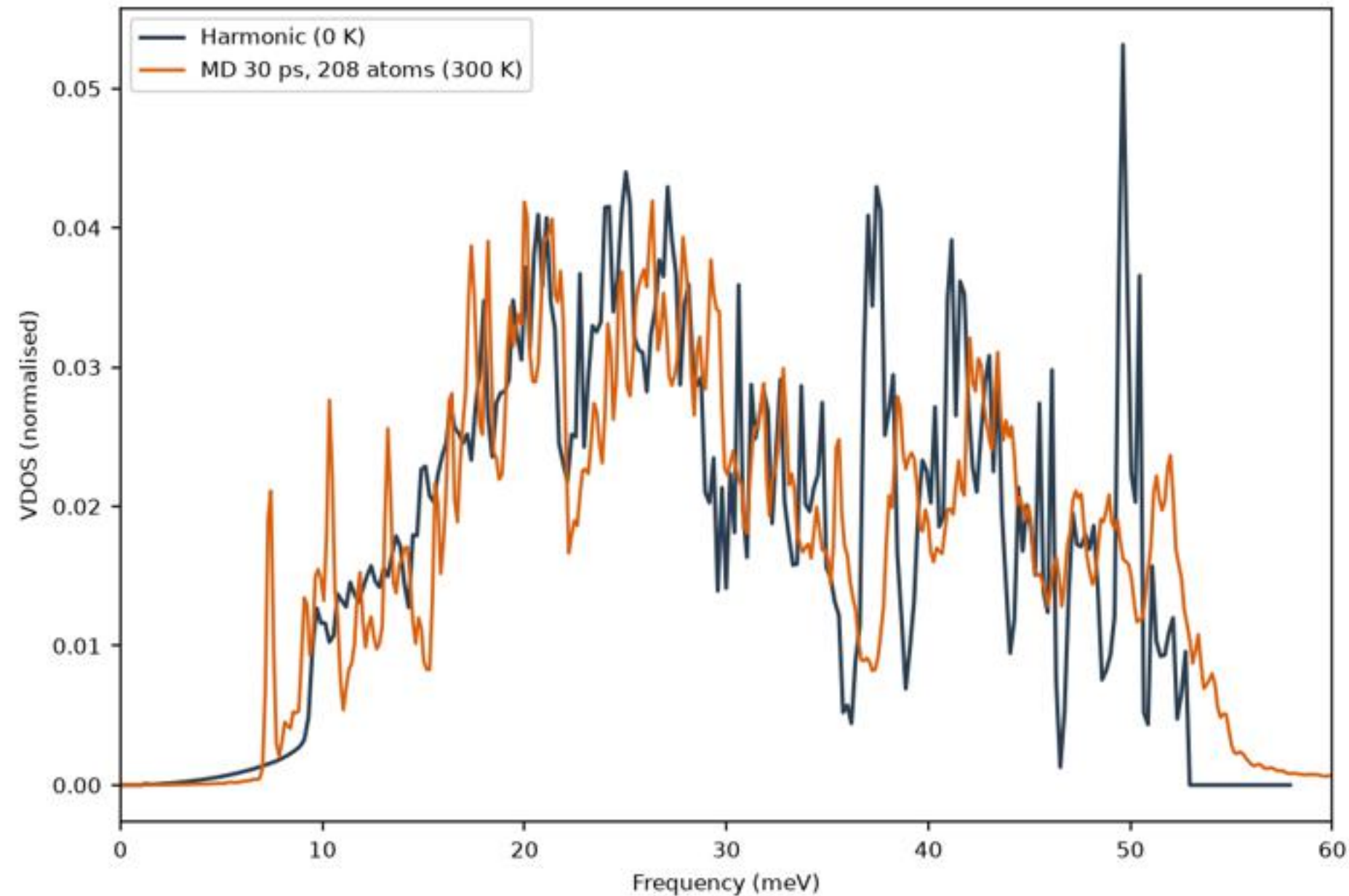
Harmonic VDOS across the size series (normalised), MACE-MPA-0 + Phonopy



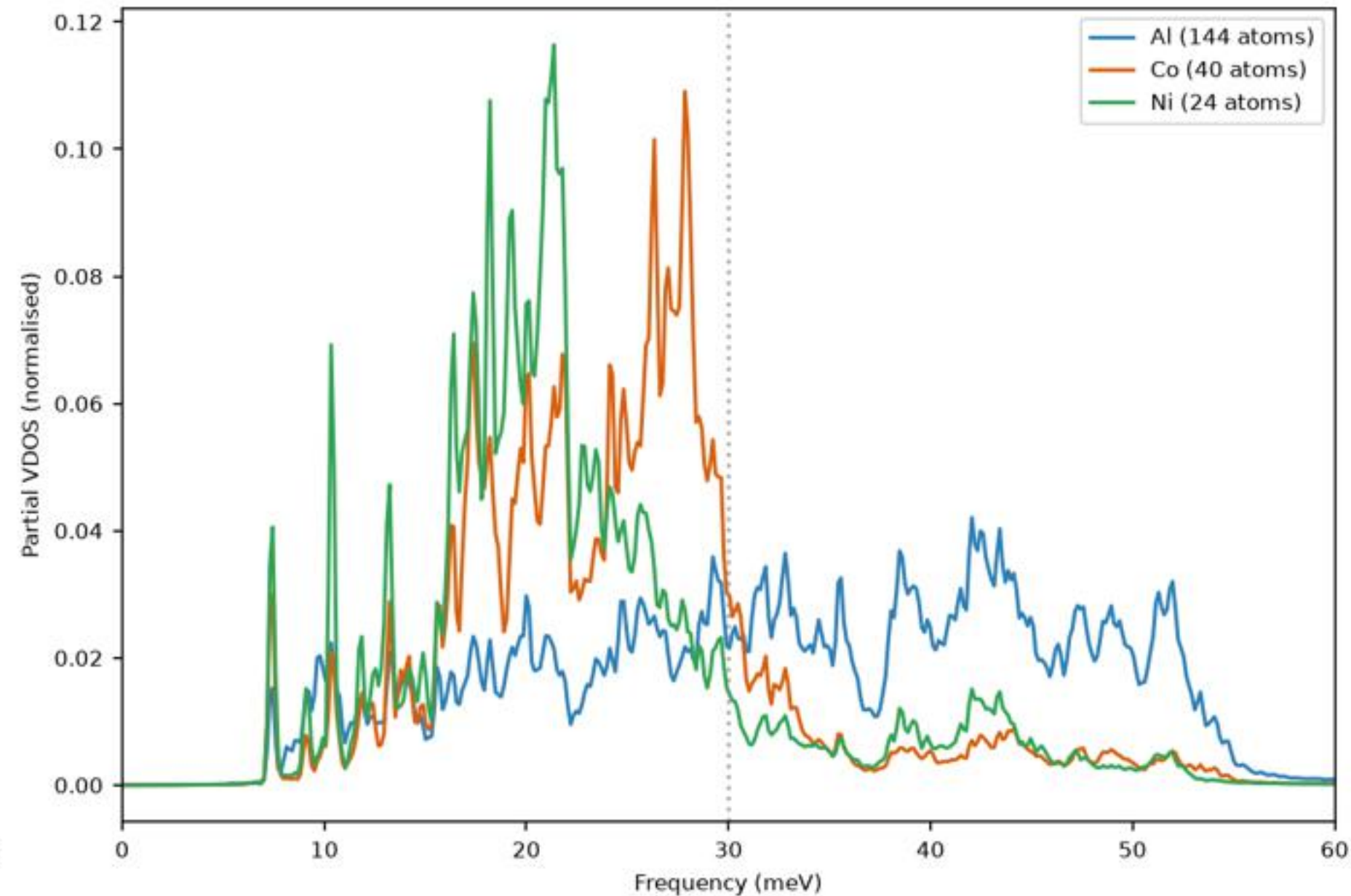
MD VDOS vs supercell, 200 ps each, 300 K



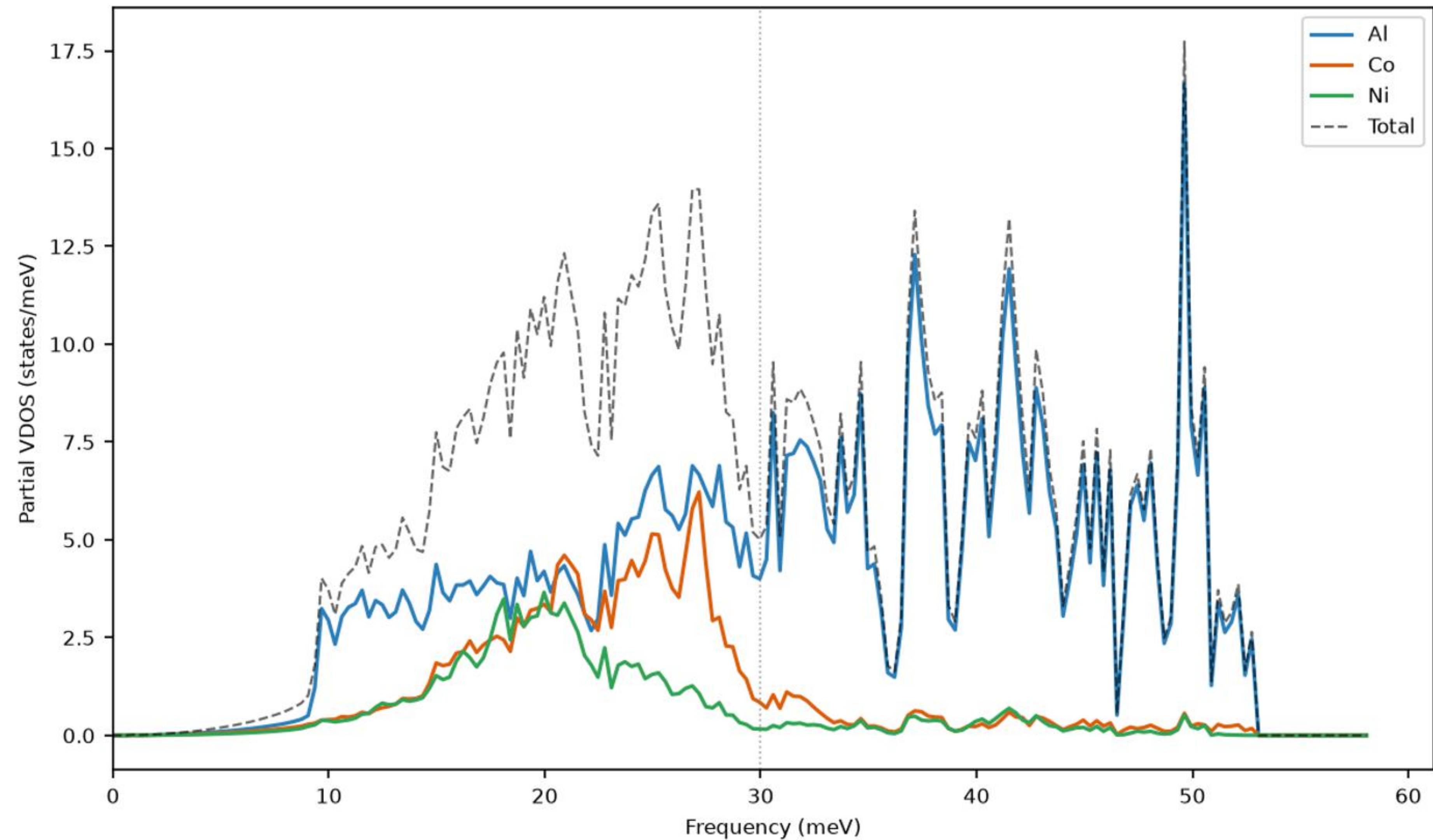
Harmonic vs anharmonic, 2x2x2 supercell



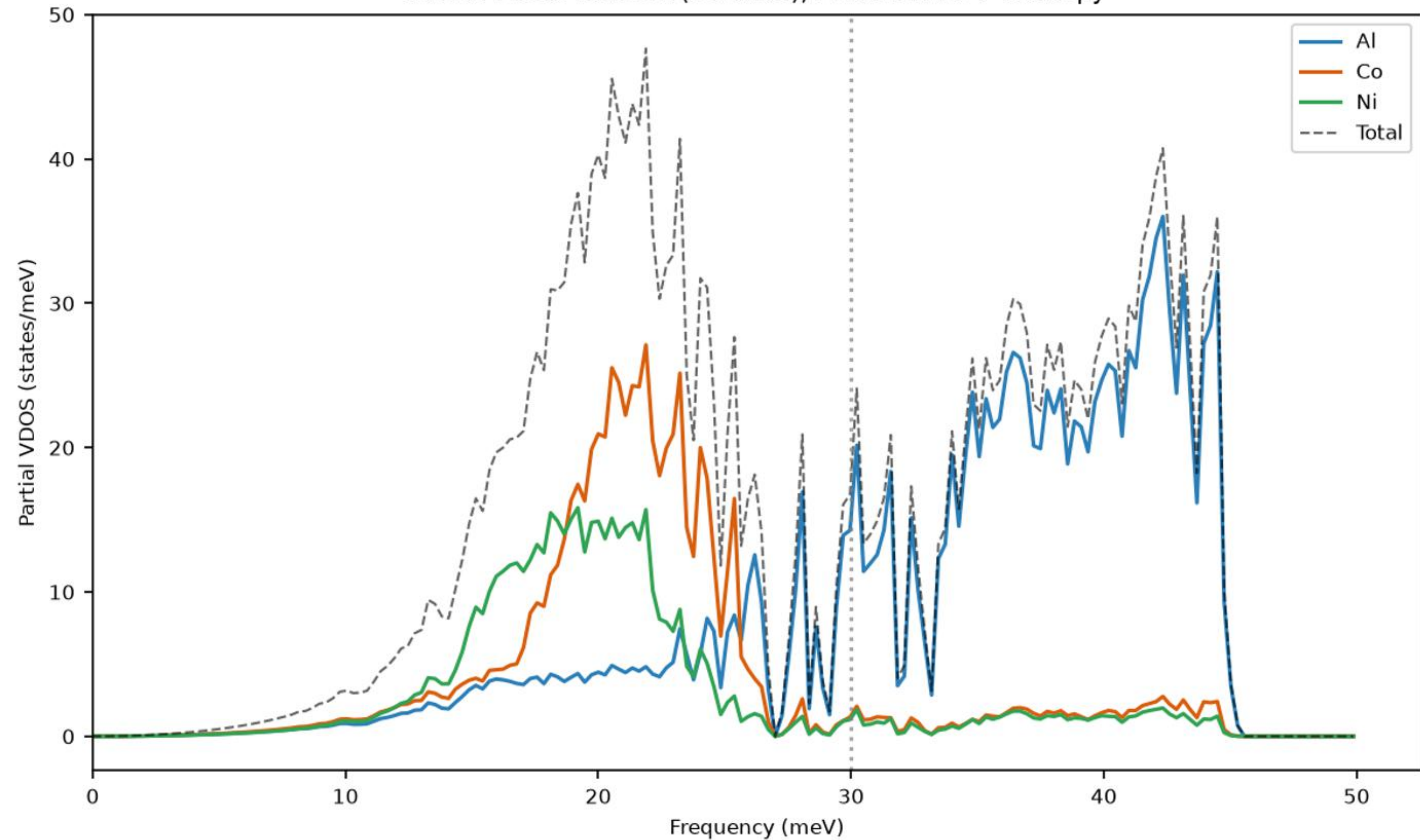
MD element decomposition, 208 atoms



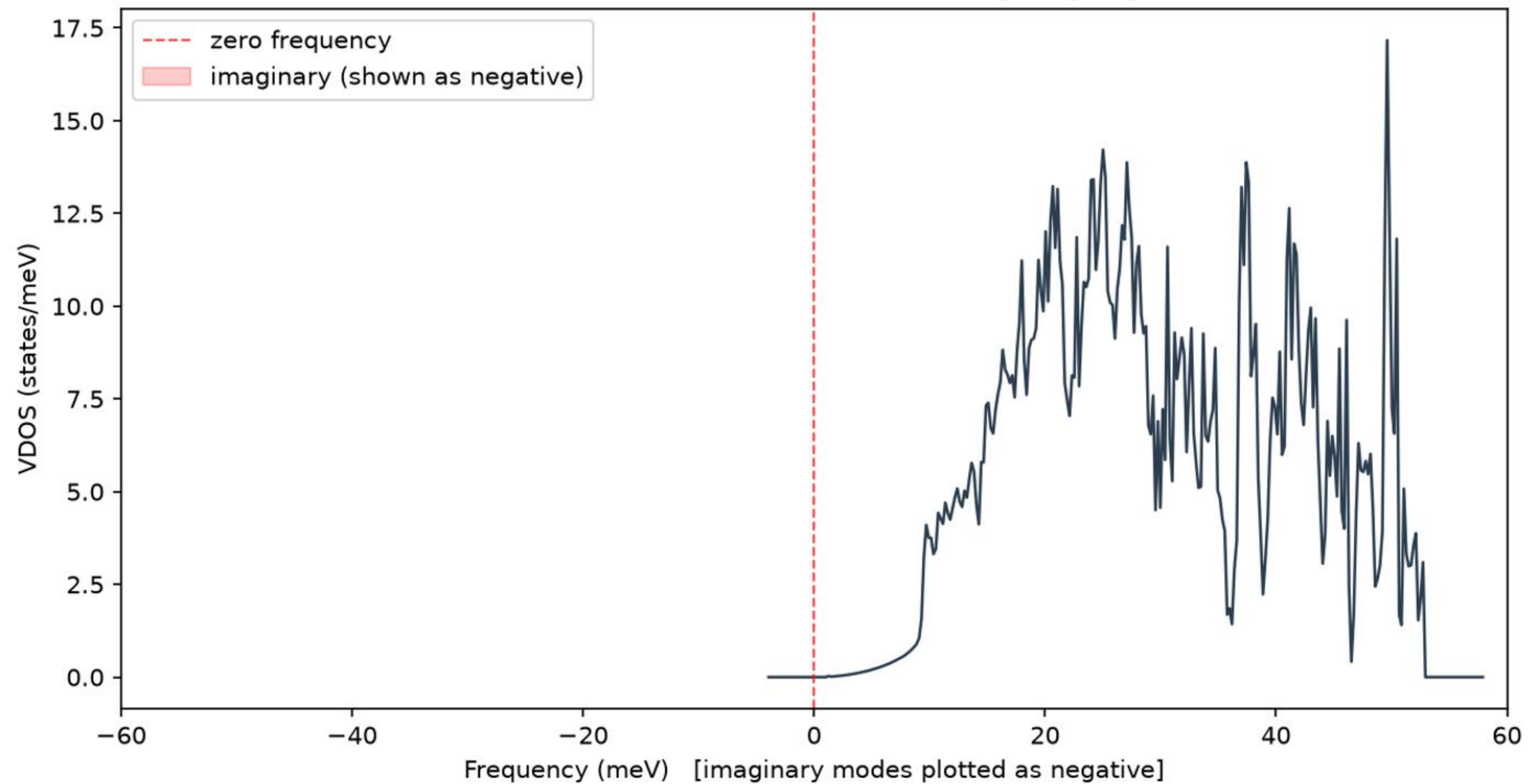
Partial VDOS: Al₉Co₂Ni₂ (26-atom), MACE-MPA-0 + Phonopy



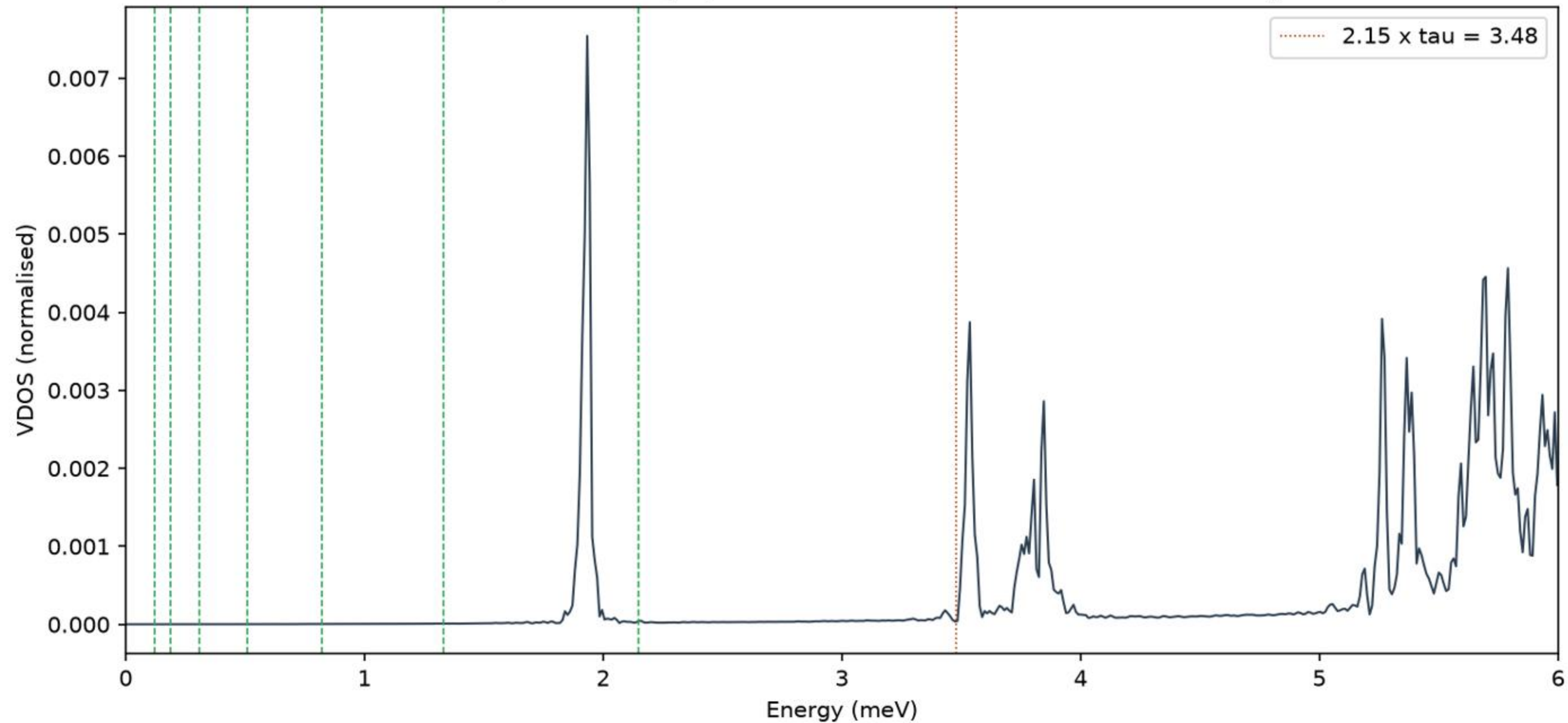
Partial VDOS: Al₂CoNi (60-atom), MACE-MPA-0 + Phonopy



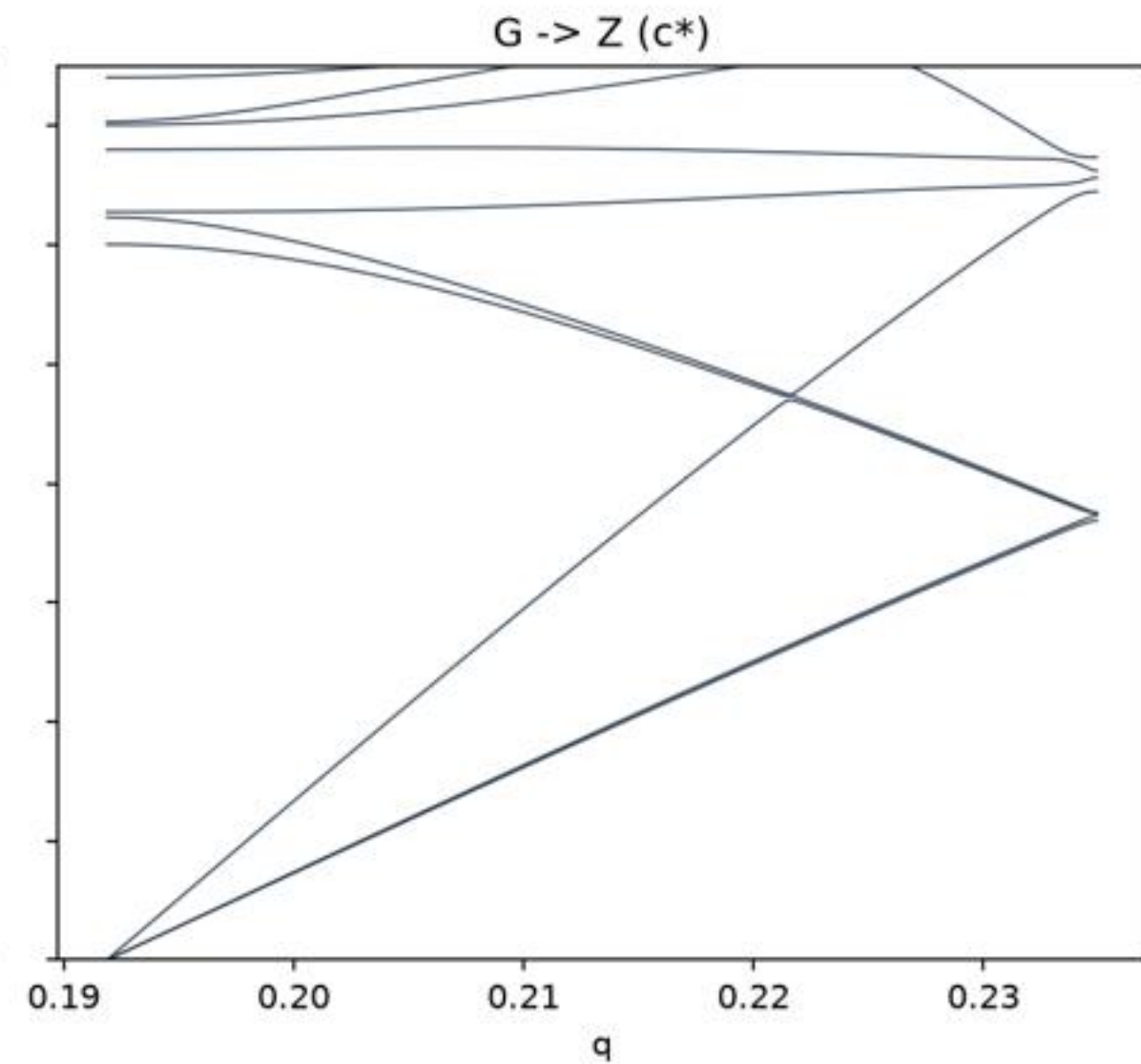
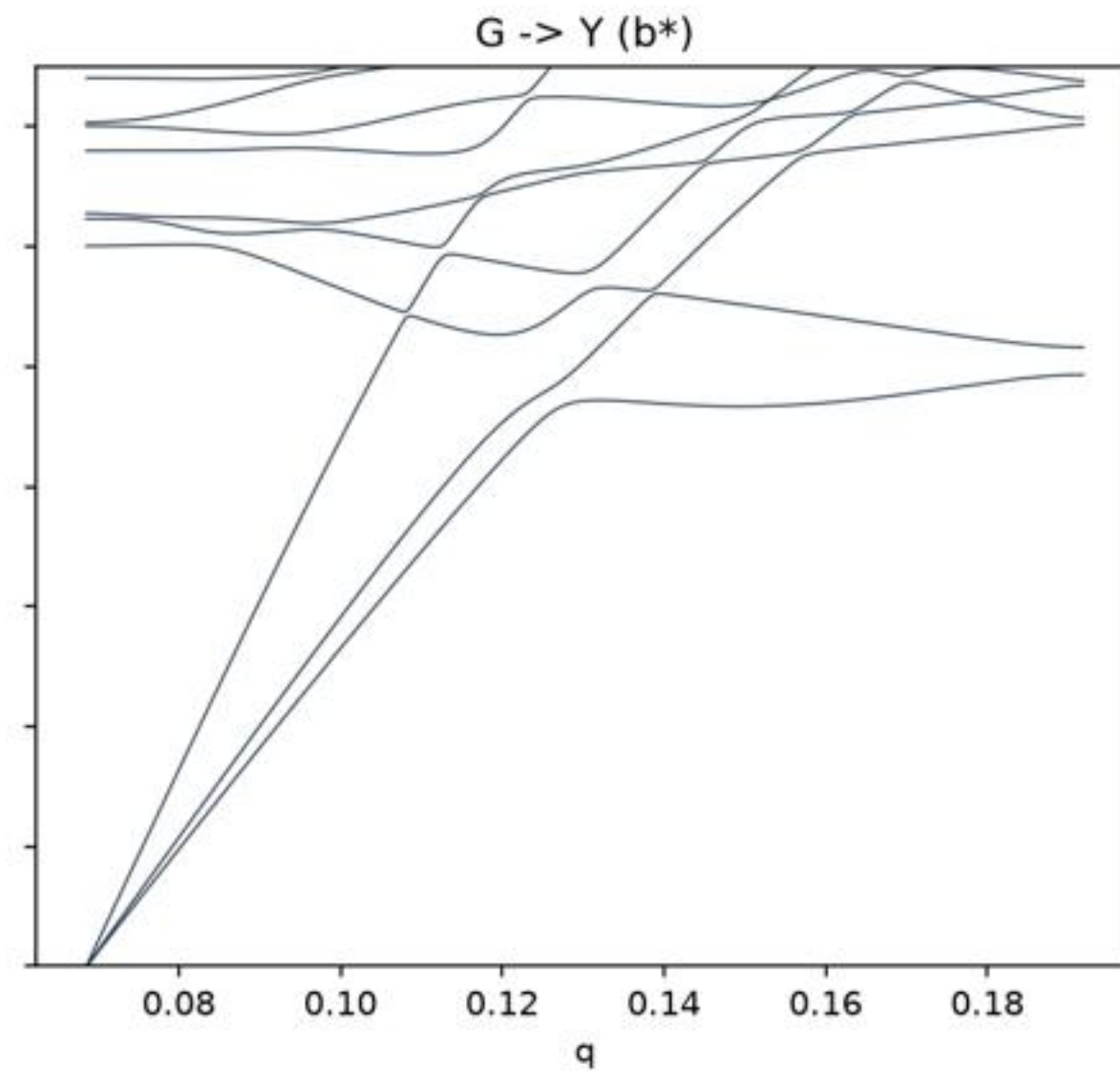
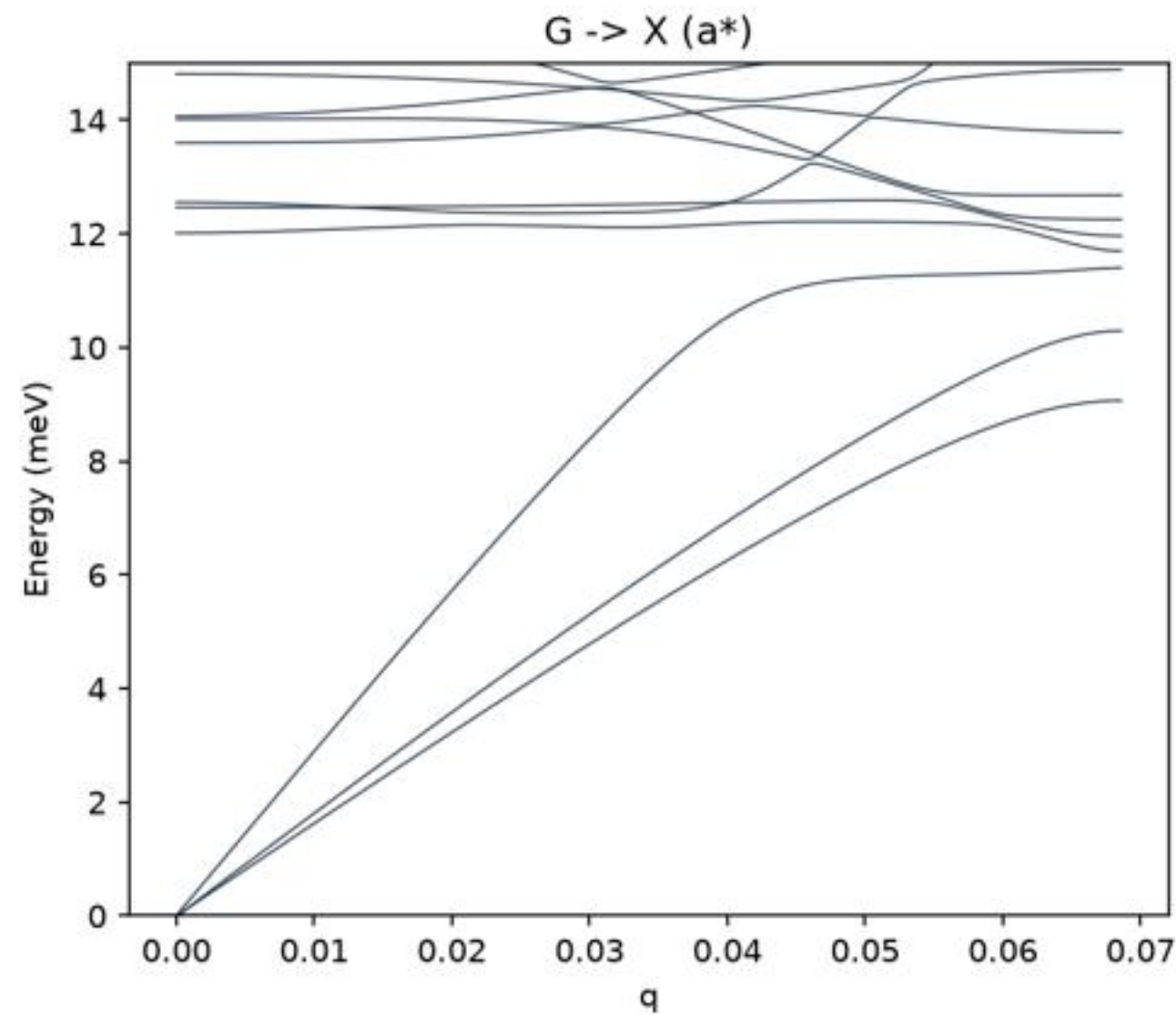
Harmonic VDOS: Al₉Co₂Ni₂ (26-atom) — imaginary region visible



8x4x4 (3328 atoms): spectrum extends below the 2.15 meV rung

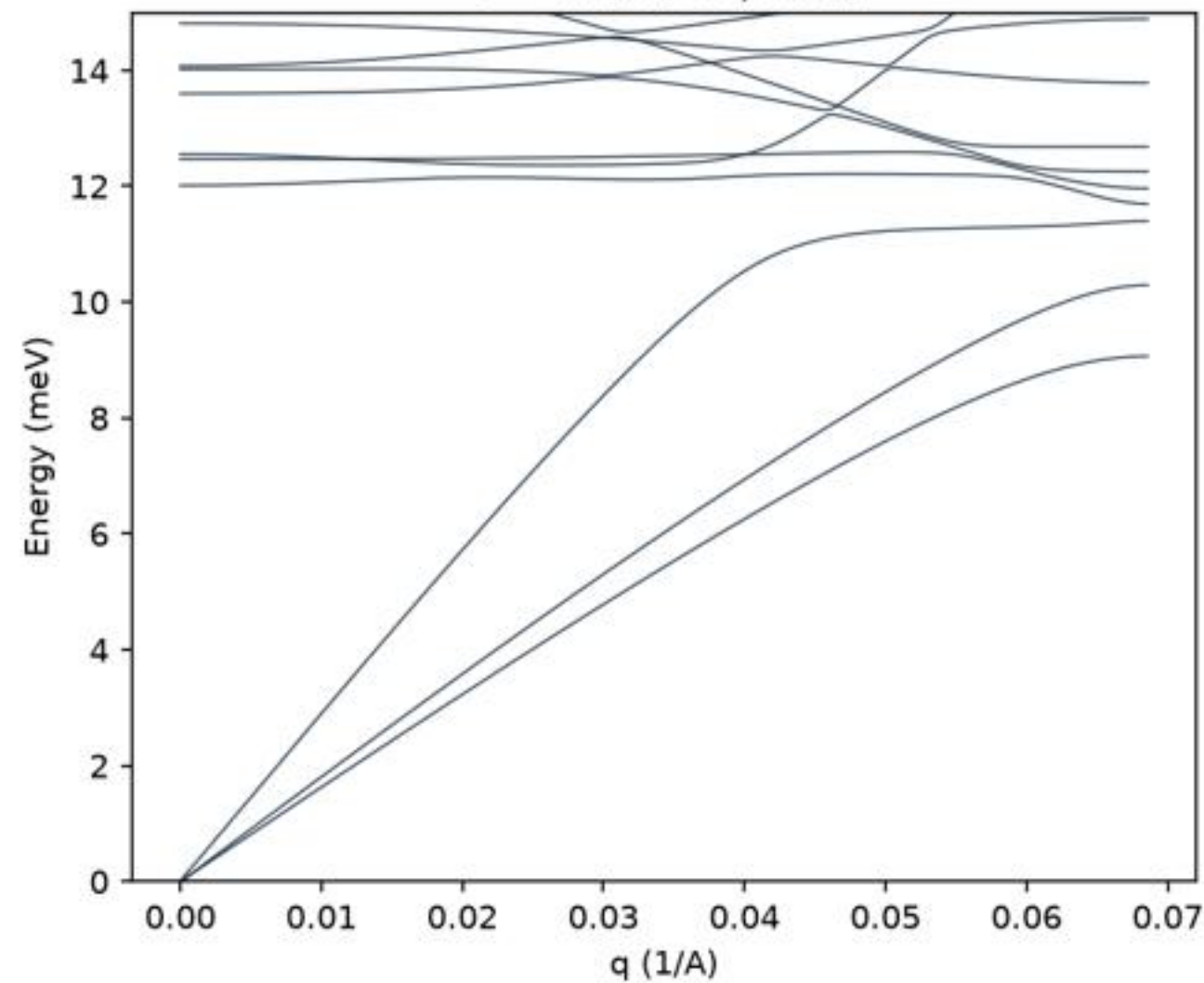


26-atom: low-frequency dispersion along the three axes

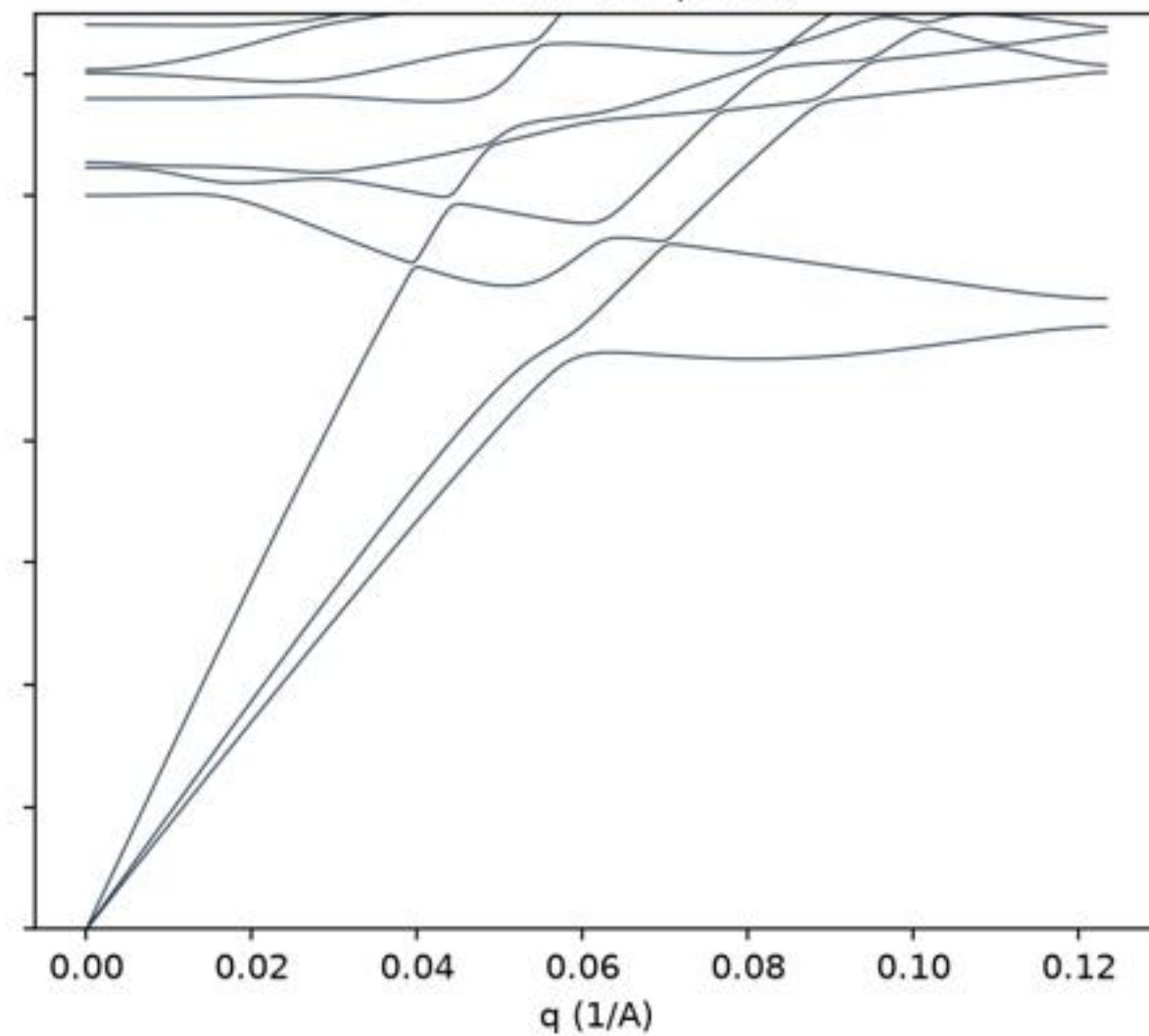


26-atom dispersion, panels rebased

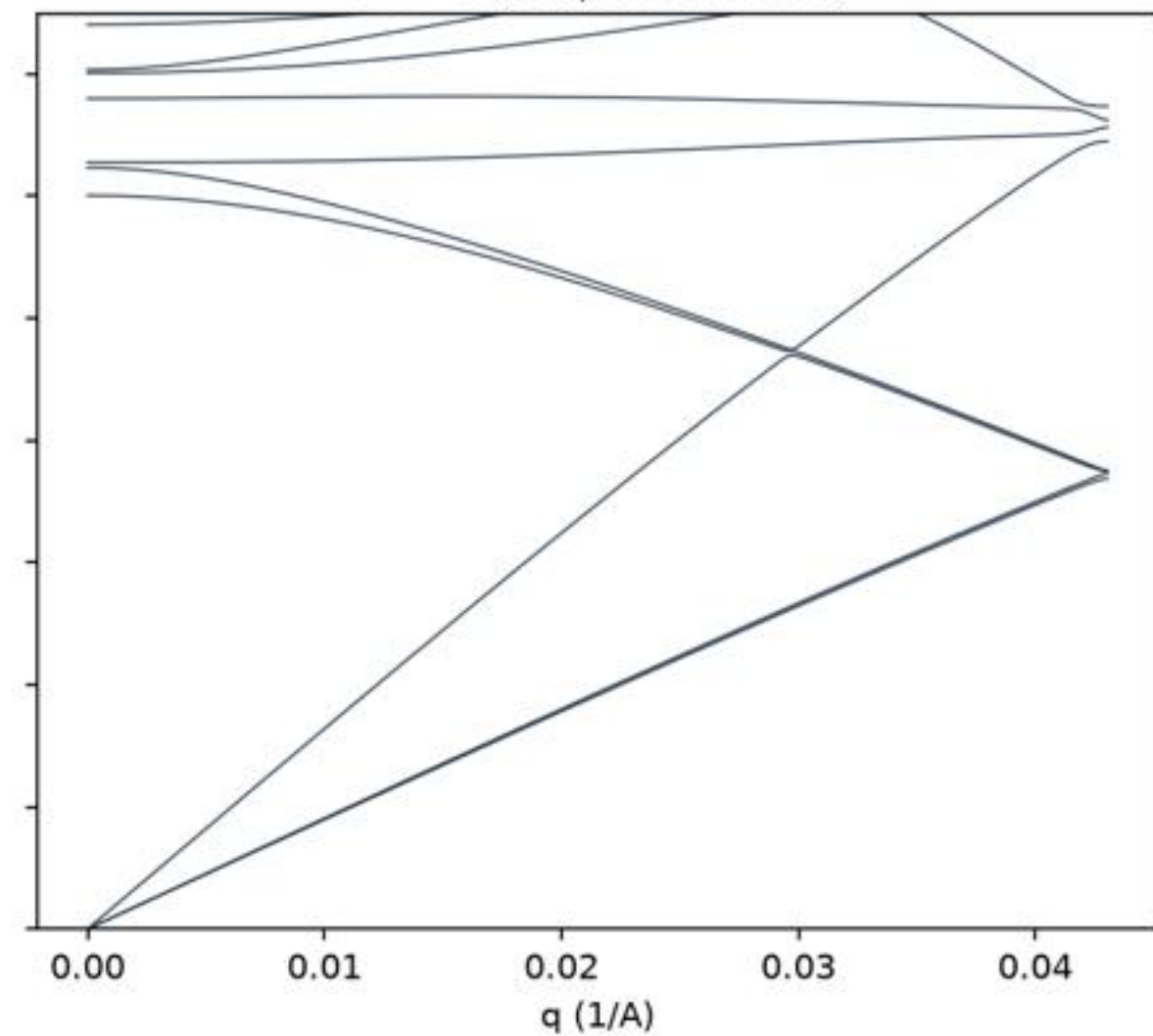
G $\rightarrow$ X (a^* , in-plane)



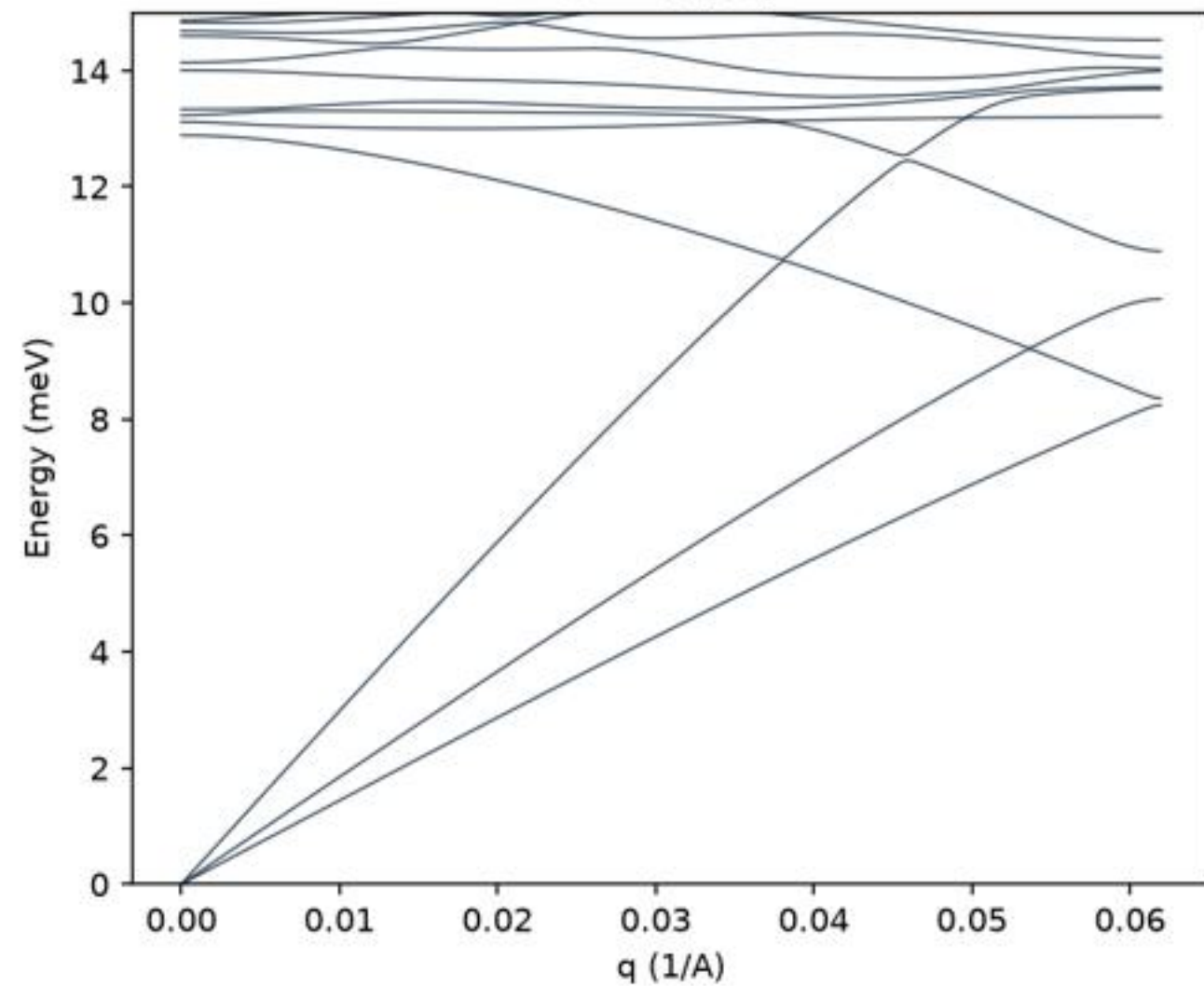
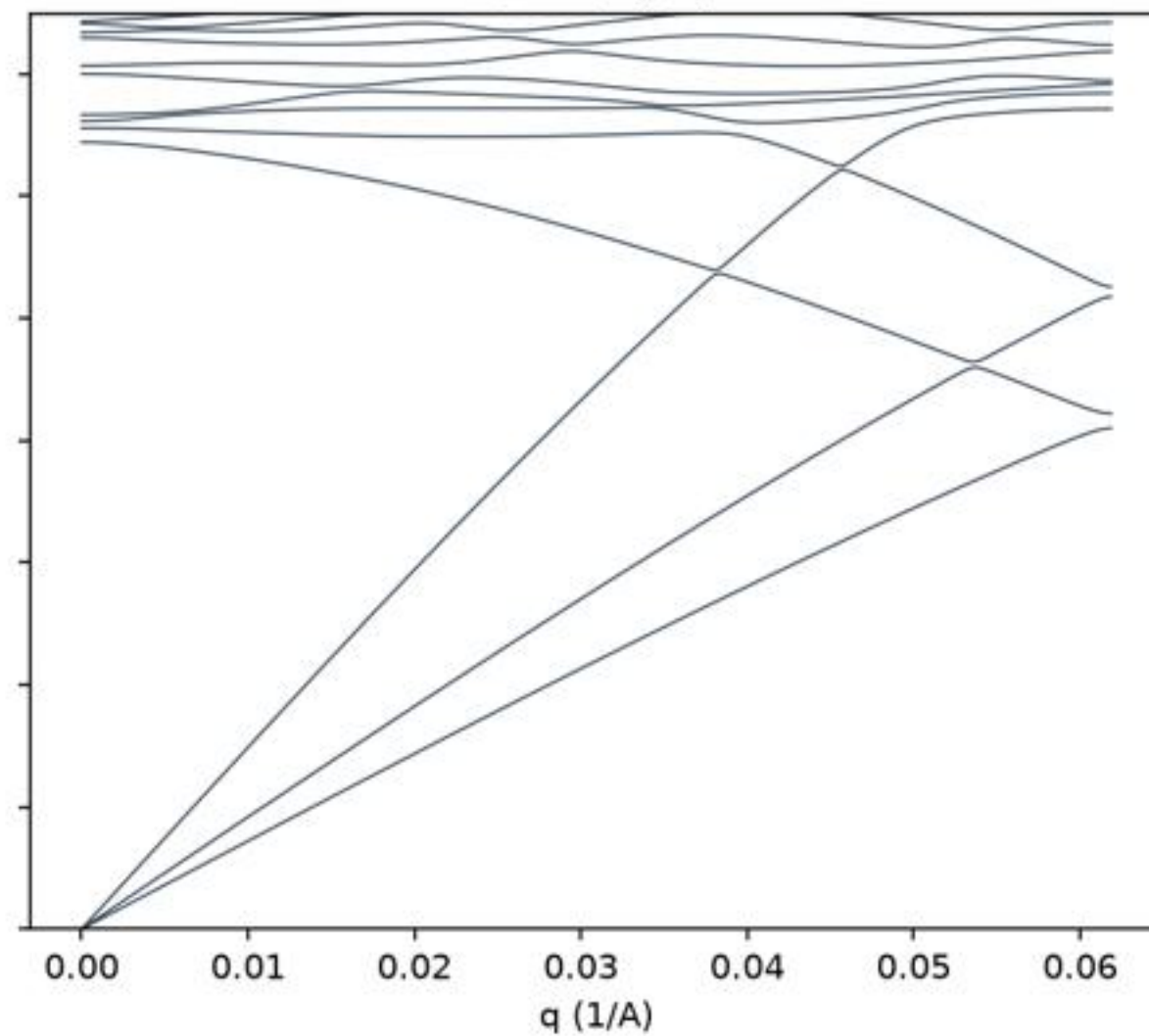
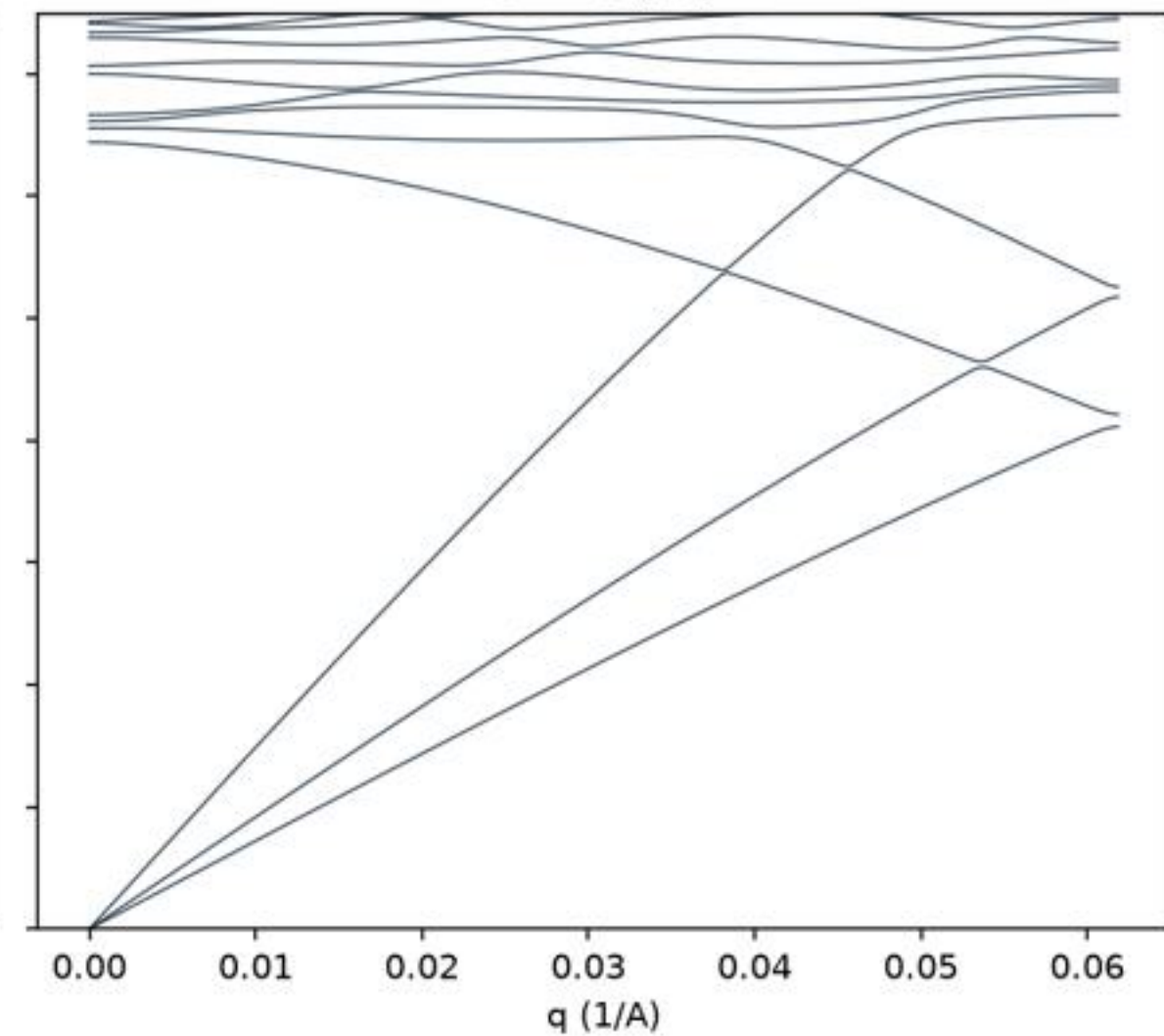
G $\rightarrow$ Y (b^* , in-plane)



G $\rightarrow$ Z (c^* , periodic axis)



60-atom dispersion

G $\rightarrow$ X (a*)G $\rightarrow$ Y (b*)G $\rightarrow$ Z (c*)

Measured finite-size floor across box shapes and structures

